Online Appendix

Subjective Annotation Studies: Valid Inference and Annotation Design

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OA.1 Pairwise reference-population functionals

This section considers reference population functionals built from two independent annotators' responses. It uses the baseline Poisson document-sampling design in Assumption **(D1)** and the uniform allocation law in Assumption **(RW1)**. Throughout, we maintain the fixed-corpus inferential framework of the main text, following the conditioning conventions in Appendix A.

Throughout, C_0 denotes a sufficiently large finite constant bound independent of J.

Recall from Lemma 2 that, conditional on $\mathcal{G}_J$, the annotator blocks $L_i = (\theta_i, (\varepsilon_{ij})_{j\le J})$ remain mutually independent and retain their original distributions. Using the same notation as in the main paper's appendix, for an annotator block $L = (\theta, (\varepsilon_j)_{j\le J})$, write

$$Y_j(L) := h(W_j, \theta, \varepsilon_j).$$

Let $\ell : \mathcal{Y}^2 \to \mathbb{R}$ be a symmetric kernel, common to all documents and specified before final-study labels are observed. Assume that its absolute value is bounded by a finite constant C_ℓ, independent of J, for all response pairs. The same reasoning applies when the kernel differs across documents, provided the document-specific kernels ℓ_j are prespecified, symmetric, and uniformly bounded by a constant independent of j and J.

For two independent reference-population annotator blocks L_1, L_2, define the document-level target and its weighted corpus average by

$$\vartheta_j := \mathbb{E}[\ell\{Y_j(L_1), Y_j(L_2)\}], \qquad \Phi_J^\ell(a) := \frac{1}{J}\sum_{j=1}^{J} a_j \vartheta_j. \tag{OA.1.1}$$

Suppose each selected document j receives $m_j \ge 2$ labels from distinct annotators. We estimate ϑ_j by the order-two U-statistic that averages the kernel over all unordered pairs of these labels:

$$\widehat{\vartheta}_j := \binom{m_j}{2}^{-1} \sum_{\substack{i,r\in M_j \\ i<r}} \ell(Y_{ij}^*, Y_{rj}^*). \tag{OA.1.2}$$

Conditional on $\mathcal{G}_J$, each pair consists of two independent reference-population responses to document j. Each kernel evaluation therefore has expectation ϑ_j, giving

$$\mathbb{E}(\widehat{\vartheta}_j \mid \mathcal{G}_J) = \vartheta_j. \tag{OA.1.3}$$

With frozen predictions g_j^ℓ of ϑ_j, define the corpus estimator

$$\widehat{\Phi}_J^\ell(a) := \frac{1}{J}\sum_{j=1}^{J} a_j g_j^\ell + \frac{1}{J}\sum_{j\in\mathcal{S}_J} \frac{a_j}{\pi_j}(\widehat{\vartheta}_j - g_j^\ell). \tag{OA.1.4}$$

Using (OA.1.3) and the inclusion probabilities in Assumption **(D1)**, iterated expectation gives

$$\mathbb{E}\{\widehat{\Phi}_J^\ell(a)\} = \frac{1}{J}\sum_{j=1}^{J} a_j g_j^\ell + \frac{1}{J}\sum_{j=1}^{J} a_j(\vartheta_j - g_j^\ell) = \Phi_J^\ell(a).$$

OA.1.1 Projection and covariance identities

For annotator blocks u and v, write

$$\widetilde{\ell}_j(u, v) := \ell\{Y_j(u), Y_j(v)\}.$$

This is the kernel evaluated at the responses that these two annotators would give to document j.

Its order-one and canonical order-two projections are

$$\gamma_{j,1}(u) := \mathbb{E}\{\widetilde{\ell}_j(u, L_2)\} - \vartheta_j, \tag{OA.1.5}$$

$$\gamma_{j,2}(u, v) := \widetilde{\ell}_j(u, v) - \vartheta_j - \gamma_{j,1}(u) - \gamma_{j,1}(v). \tag{OA.1.6}$$

These are the usual Hoeffding projections (van der Vaart, 1998, Ch. 12). They are uniformly bounded and satisfy

$$\mathbb{E}\{\gamma_{j,1}(L_1)\} = 0, \qquad \mathbb{E}\{\gamma_{j,2}(L_1, L_2) \mid L_s\} = 0, \quad s = 1, 2. \tag{OA.1.7}$$

Define

$$\sigma^2_{j,s} := \mathbb{E}\{\gamma_{j,s}(L_{1:s})^2\}, \qquad s = 1, 2, \tag{OA.1.8}$$

$$\tau_{jk,1} := \mathbb{E}\{\gamma_{j,1}(L_1)\gamma_{k,1}(L_1)\}, \tag{OA.1.9}$$

$$\tau_{jk,2} := \mathbb{E}\{\gamma_{j,2}(L_1, L_2)\gamma_{k,2}(L_1, L_2)\}. \tag{OA.1.10}$$

Lemma OA.1 (Order-two projection and covariance formulas)**.** *For every selected document j,*

$$\widehat{\vartheta}_j - \vartheta_j = \frac{2}{m_j} \sum_{i \in M_j} \gamma_{j,1}(L_i) + \binom{m_j}{2}^{-1} \sum_{\substack{i,r \in M_j \\ i<r}} \gamma_{j,2}(L_i, L_r). \tag{OA.1.11}$$

Consequently,

$$\operatorname{Var}(\widehat{\vartheta}_j \mid \mathcal{G}_J) = \frac{4}{m_j}\sigma^2_{j,1} + \binom{m_j}{2}^{-1} \sigma^2_{j,2}. \tag{OA.1.12}$$

For distinct selected documents $j \neq k$,

$$\begin{aligned} &\operatorname{Cov}(\widehat{\vartheta}_j, \widehat{\vartheta}_k \mid \mathcal{G}_J) \\ &\quad = \frac{4N_{jk}}{m_j m_k}\tau_{jk,1} + \frac{\binom{N_{jk}}{2}}{\binom{m_j}{2}\binom{m_k}{2}}\tau_{jk,2}, \qquad N_{jk} := |M_j \cap M_k|. \end{aligned} \tag{OA.1.13}$$

Proof. Substitute

$$\widetilde{\ell}_j(u, v) = \vartheta_j + \gamma_{j,1}(u) + \gamma_{j,1}(v) + \gamma_{j,2}(u, v)$$

into (OA.1.2). Each annotator appears in $m_j - 1$ unordered pairs, so the coefficient on each first-order term is

$$\frac{m_j - 1}{\binom{m_j}{2}} = \frac{2}{m_j}.$$

This gives (OA.1.11).

Conditional on $\mathcal{G}_J$, the annotator blocks are independent. Together with the zero conditional mean property in (OA.1.7), this implies that first- and second-order terms have zero covariance. Within each order, terms involving different annotators or different unordered annotator pairs also have zero covariance. Summing the individual variances therefore gives (OA.1.12).

Across documents j and k, only first-order terms involving the same annotator and second-order terms involving the same pair of annotators can contribute to the covariance. There are N_{jk} such annotators and $\binom{N_{jk}}{2}$ such pairs, contributing $\tau_{jk,1}$ and $\tau_{jk,2}$, respectively. Multiplying by the coefficients in (OA.1.11) gives (OA.1.13). □

Let $r^\ell_j := \vartheta_j - g^\ell_j$. Adding and subtracting ϑ_j in (OA.1.4) gives

$$\widehat{\Phi}^\ell_J(a) - \Phi^\ell_J(a) = T^s_{J,\ell}(a) + T^m_{J,\ell}(a), \tag{OA.1.14}$$

$$T^s_{J,\ell}(a) := \frac{1}{J}\sum_{j=1}^{J} \left(\frac{I_j}{\pi_j} - 1\right) a_j r^\ell_j,$$

$$T^m_{J,\ell}(a) := \frac{1}{J}\sum_{j \in \mathcal{S}_J} \frac{a_j}{\pi_j}(\widehat{\vartheta}_j - \vartheta_j).$$

Lemma OA.1 yields the scaled measurement variance conditional on the allocation graph

$$\begin{aligned} v^{\ell}_{\varepsilon,J}(a) &:= \bar{B}\,\mathrm{Var}\{T^{m}_{J,\ell}(a) \mid \mathcal{G}_J\} \\ &= \frac{\bar{B}}{J^2}\sum_{j=1}^{J}\frac{I_j a_j^2}{\pi_j^2}\left\{\frac{4}{m_j}\sigma^2_{j,1} + \frac{\sigma^2_{j,2}}{\binom{m_j}{2}}\right\} \\ &\quad + \frac{\bar{B}}{J^2}\sum_{j\neq k}\frac{I_j I_k a_j a_k}{\pi_j \pi_k}\left\{\frac{4N_{jk}}{m_j m_k}\tau_{jk,1} + \frac{\binom{N_{jk}}{2}}{\binom{m_j}{2}\binom{m_k}{2}}\tau_{jk,2}\right\}, \end{aligned} \tag{OA.1.15}$$

where the off-diagonal sum is over ordered document pairs.

OA.1.2 Variance stabilization and the CLT

Equation (OA.1.15) separates the conditional measurement variance into three terms: the within-document variance, the cross-document covariances from the first- and second-order projections. Our goal is to show that, under regularity conditions similar to those imposed in the main text, the within-document variance contribution,

$$D^{\ell}_J(a) := \frac{\bar{B}}{J^2}\sum_{j=1}^{J}\frac{I_j a_j^2}{\pi_j^2}\left(\frac{4\sigma^2_{j,1}}{m_j} + \frac{\sigma^2_{j,2}}{\binom{m_j}{2}}\right),$$

and the first-order cross-document covariance contribution,

$$C^{\ell}_{1,J}(a) := \frac{4\bar{B}}{J^2}\sum_{j\neq k}\frac{I_j I_k N_{jk}}{\pi_j \pi_k m_j m_k} a_j a_k \tau_{jk,1},$$

converge to deterministic limits, while the second-order cross-document covariance contribution,

$$C^{\ell}_{2,J}(a) := \frac{\bar{B}}{J^2}\sum_{j\neq k}\frac{I_j I_k \binom{N_{jk}}{2}}{\pi_j \pi_k \binom{m_j}{2}\binom{m_k}{2}} a_j a_k \tau_{jk,2},$$

vanishes.

To write the within-document term in the same form as in the weighted-mean case, define

$$d^{\ell}_j := m_j\,\mathrm{Var}(\widehat{\vartheta}_j \mid \mathcal{G}_J) = 4\sigma^2_{j,1} + \frac{2}{m_j - 1}\sigma^2_{j,2}. \tag{OA.1.16}$$

Thus the document-level variance is d^{ℓ}_j/m_j, with d^{ℓ}_j playing the role of ν_j in the weighted-mean case. It includes both projection orders: only the second-order *cross-document covariance* will be negligible.

Under the fixed-corpus convention, the projection variances and prespecified annotation depths are deterministic. Hence d^{ℓ}_j is deterministic, and boundedness of ℓ, together with $m_j \geq 2$, makes it uniformly bounded.

The diagonal contribution can therefore be written as

$$D^{\ell}_J(a) := \frac{\bar{B}}{J^2}\sum_{j=1}^{J}\frac{I_j a_j^2 d^{\ell}_j}{\pi_j^2 m_j} = \frac{1}{J}\sum_{j=1}^{J}\frac{I_j}{\pi_j}\frac{a_j^2 d^{\ell}_j}{\lambda_j}.$$

This is a Horvitz–Thompson average, so its concentration can be established by the same argument as in the weighted-mean case. The two cross-document terms additionally require control of annotator overlaps.

The first-order covariance term has the exact same form as the overlap term in the weighted average case, with τ_{jk} replaced by $4\tau_{jk,1}$. As a result, under the same uniform allocation law, the concentration argument in the proof of Lemma 8 applies to arbitrary fixed coefficients. The following lemma records this consequence of the main text proof which we can re-use:

Lemma OA.2 (Concentration of weighted annotator overlaps)**.** *Consider selected document and annotator degree sequences with common degree sum $B \to \infty$ and uniformly bounded degrees on both sides. Suppose that these sequences admit at least one simple bipartite graph, and let $\mathcal{A}$ be uniform over all labeled simple bipartite graphs with these degrees.*

Let N_{jk} denote the number of annotators assigned to both documents j and k. For any deterministic real coefficients $\{c_{jk} : j \neq k\}$, indexed by ordered pairs of distinct documents,

$$\operatorname{Var}\left(\sum_{j\neq k} c_{jk} N_{jk}\right) \leq CB \max_{j\neq k} |c_{jk}|^2 \tag{OA.1.17}$$

for all sufficiently large B, where $C < \infty$ depends only on the degree bounds. The coefficients may vary along the sequence.

The same bound holds for the conditional variance if, under the conditioning, the degrees and coefficients are fixed and $\mathcal{A}$ remains uniform over all simple bipartite graphs with those degrees.

Proof. For each active annotator i, let $\mathcal{J}_i$ be the set of assigned documents and define

$$U_i(c) := \sum_{\substack{j,k\in\mathcal{J}_i \\ j\neq k}} c_{jk}, \qquad M := \max_{j\neq k} |c_{jk}|.$$

Counting each annotator shared by an ordered document pair once gives

$$\sum_{j\neq k} c_{jk} N_{jk} = \sum_i U_i(c).$$

Let K be a fixed upper bound on annotator degrees. Since $|U_i(c)| \leq K(K-1)M$, we have

$$\operatorname{Var}\{U_i(c)\} \leq C_0 M^2.$$

For distinct annotators, apply the comparison between the joint distribution of their assigned document sets and the product of their marginal distributions in (C.21). Applied to $U_i(c)U_r(c)$, whose absolute value is bounded by $K^2(K-1)^2M^2$, this gives

$$|\operatorname{Cov}\{U_i(c), U_r(c)\}| \leq \frac{C_0 M^2}{B}, \qquad i \neq r,$$

where $C_0 < \infty$ depends only on the degree bounds.

There are $n \leq B$ active annotators, since each has at least one assignment. Therefore,

$$\begin{aligned}
\operatorname{Var}\left(\sum_{j\neq k} c_{jk} N_{jk}\right) &= \operatorname{Var}\left(\sum_{i=1}^{n} U_i(c)\right) \\
&\leq \sum_{i=1}^{n} \operatorname{Var}\{U_i(c)\} + \sum_{i\neq r} |\operatorname{Cov}\{U_i(c), U_r(c)\}| \\
&\leq C_0 M^2 \left\{ n + \frac{n(n-1)}{B} \right\} \\
&\leq 2C_0 B M^2.
\end{aligned}$$

Taking $C = 2C_0$ proves (OA.1.17). The same argument applies under the stated conditional law. □

Lemma OA.3 (Limit of weighted annotator overlaps)**.** *Suppose Assumptions **(D1)** and **(RW1)** hold, with fixed annotator capacity K, and $\bar{B} \to \infty$. Suppose that, for constants $\bar{m} < \infty$ and $c_\lambda > 0$,*

$$2 \le m_j \le \bar{m}, \qquad \lambda_j := \frac{J\pi_j m_j}{\bar{B}} \ge c_\lambda$$

uniformly in j and J. Let $\{b_{jk} : j \ne k\}$ be a deterministic symmetric array satisfying $\max_{j\ne k} |b_{jk}| \le C_b$, where $C_b < \infty$ is independent of J, and define

$$W_J(b) := \frac{\bar{B}}{J^2} \sum_{j\ne k} \frac{I_j I_k N_{jk}}{\pi_j \pi_k m_j m_k} b_{jk}.$$

If $J^{-2}\sum_{j\ne k} b_{jk} \to t_b \in \mathbb{R}$, then

$$W_J(b) \xrightarrow{p} (K-1)t_b. \tag{OA.1.18}$$

Proof. The off-diagonal term $C_J(a)$ in the proof of Theorem 2 has exactly the form $W_J(b)$, with $b_{jk} = a_j a_k \tau_{jk}$. We apply that argument to the array b_{jk} specified here.

The coefficient on each sampled overlap satisfies

$$\left| \frac{\bar{B}\, b_{jk}}{J^2 \pi_j \pi_k m_j m_k} \right| = \frac{|b_{jk}|}{\bar{B}\lambda_j \lambda_k} \le \frac{C_b}{c_\lambda^2 \bar{B}}.$$

This is the bound used in the earlier proof to control allocation fluctuations and the error in approximating expected overlaps. The document-sampling calculation in (C.23) and its subsequent bounds likewise apply, because they use only the determinism, symmetry, and uniform boundedness of b_{jk}. The budget concentration and allocation feasibility conclusions used there also hold under the present assumptions. The same argument therefore gives

$$W_J(b) = \frac{K-1}{J^2} \sum_{j\ne k} b_{jk} + o_p(1).$$

The assumed limit of the deterministic average proves (OA.1.18). □

Lemma OA.4 (Vanishing second-order cross-document covariance)**.** *Maintain the sampling and allocation conditions of Lemma OA.3, together with the boundedness conditions on the target weights a and kernel ℓ. Then*

$$C^{\ell}_{2,J}(a) = o_p(1). \tag{OA.1.19}$$

Proof. Let

$$H_{2,J} := \sum_{\substack{j<k \\ j,k\in\mathcal{S}_J}} \binom{N_{jk}}{2}.$$

This counts choices of two documents and two annotators such that both annotators evaluate both documents. We first show that this count is bounded in probability.

Recall the event $\mathcal{H}_J$ from (C.12), on which the capacity-filled allocation is feasible and $\bar{B}/2 \le B \le 2\bar{B}$. This event is determined by $\mathcal{S}_J$, and (C.13) gives $\mathbb{P}(\mathcal{H}_J^c) \to 0$.

For distinct selected documents j, k and distinct active annotators i, r, the event $i, r \in M_j \cap M_k$ requires the four assignments $(j,i), (j,r), (k,i), (k,r)$. On $\mathcal{H}_J$, applying Lemma 7 to these four assignments and using the bounded degrees gives

$$\mathbb{P}(i, r \in M_j \cap M_k \mid \mathcal{S}_J) \le \frac{C_0}{B^4},$$

uniformly for sufficiently large J. Here $C_0 < \infty$ is a sufficiently large constant depending only on the fixed bounds.

Each contribution to $H_{2,J}$ consists of two documents and two annotators, with both annotators assigned to both documents. There are $\binom{|\mathcal{S}_J|}{2}$ ways to choose the documents. For each such choice, there are $\binom{n}{2}$ ways to choose the annotators, where n is the number of active annotators.

Each choice of a document pair and an annotator pair contributes to $H_{2,J}$ only when all four required assignments are present. The preceding bound shows that this event has conditional probability at most C_0/B^4. Summing over all possible choices therefore gives, on $\mathcal{H}_J$,

$$\mathbb{E}(H_{2,J} \mid \mathcal{S}_J) \le \binom{|\mathcal{S}_J|}{2}\binom{n}{2}\frac{C_0}{B^4} \le C_0.$$

For the last inequality, each selected document and each active annotator accounts for at least one of the B assignments. Thus $|\mathcal{S}_J| \le B$ and $n \le B$, so there are at most B^4 choices, each with probability at most C_0/B^4.

This conditional expectation bound holds on $\mathcal{H}_J$. Since $\mathcal{H}_J$ is determined by $\mathcal{S}_J$, conditional Markov's inequality gives, for every $M > 0$,

$$\begin{aligned}\mathbb{P}(H_{2,J} > M) &\le \mathbb{P}(\mathcal{H}_J^c) + \mathbb{E}[\mathbf{1}_{\mathcal{H}_J}\mathbb{P}(H_{2,J} > M \mid \mathcal{S}_J)] \\ &\le \mathbb{P}(\mathcal{H}_J^c) + \frac{1}{M}\mathbb{E}[\mathbf{1}_{\mathcal{H}_J}\mathbb{E}(H_{2,J} \mid \mathcal{S}_J)] \\ &\le \mathbb{P}(\mathcal{H}_J^c) + \frac{C_0}{M}.\end{aligned}$$

Since $\mathbb{P}(\mathcal{H}_J^c) \to 0$, this proves

$$H_{2,J} = O_p(1). \tag{OA.1.20}$$

We next bound the coefficient of each shared annotator pair. Boundedness of the kernel implies that $\tau_{jk,2}$ is uniformly bounded. Using $\lambda_j = J\pi_j m_j/\bar{B}$, we obtain

$$\frac{\bar{B}|a_j a_k \tau_{jk,2}|}{J^2\pi_j\pi_k\binom{m_j}{2}\binom{m_k}{2}} = \frac{4|a_j a_k \tau_{jk,2}|}{\bar{B}\lambda_j\lambda_k(m_j-1)(m_k-1)} \le \frac{C_0}{\bar{B}}.$$

The sum defining $C_{2,J}^{\ell}(a)$ uses ordered document pairs, so its total shared-pair count is $2H_{2,J}$. Hence

$$|C_{2,J}^{\ell}(a)| \le \frac{2C_0}{\bar{B}}H_{2,J} = o_p(1).$$

□

Theorem OA.1 (Stabilization for pairwise functionals)**.** *Maintain the response and design assumptions of this section and the conditions of Lemma OA.4. Suppose the deterministic averages satisfy*

$$\frac{1}{J}\sum_{j=1}^{J}\frac{a_j^2 d_j^{\ell}}{\lambda_j} \longrightarrow v_{\mathrm{diag}}^{\ell}(a), \tag{OA.1.21}$$

$$\frac{1}{J^2}\sum_{j\neq k} a_j a_k \tau_{jk,1} \longrightarrow \tau_1^{\ell}(a), \tag{OA.1.22}$$

with finite limits. Then

$$v_{\varepsilon,J}^{\ell}(a) \xrightarrow{p} v_{\mathrm{diag}}^{\ell}(a) + 4(K-1)\tau_1^{\ell}(a). \tag{OA.1.23}$$

Proof. Since d_j^{ℓ} is deterministic and uniformly bounded, the diagonal argument in the proof of Theorem 2 applies with ν_j replaced by d_j^{ℓ}. Using (OA.1.21), we obtain

$$D_J^{\ell}(a) = v_{\mathrm{diag}}^{\ell}(a) + o_p(1). \tag{OA.1.24}$$

For the first-order cross-document covariance, $C^{\ell}_{1,J}(a) = W_J(b)$ with $b_{jk} = 4a_j a_k \tau_{jk,1}$. This array is deterministic, symmetric, and uniformly bounded, so Lemma OA.3 and (OA.1.22) give

$$C^{\ell}_{1,J}(a) = 4(K-1)\tau_1^{\ell}(a) + o_p(1). \tag{OA.1.25}$$

Lemma OA.4 gives $C^{\ell}_{2,J}(a) = o_p(1)$. Adding these three components of (OA.1.15) proves the result. □

We next establish a central limit theorem for the pairwise estimator. The proof follows the same decomposition as in the weighted-mean case. The document-sampling term is covered by Lemma 1, with r_j replaced by r_j^{ℓ}. We therefore focus on the Gaussian approximation for the measurement term conditional on $\mathcal{G}_J$, using the variance stabilization established above. Once this approximation is obtained, the two terms are combined by the same conditional characteristic-function argument used in the proof of Theorem 1.

Theorem OA.2 (CLT for pairwise functionals)**.** *Suppose the conditions of Theorem OA.1 hold, the frozen predictions g_j^{ℓ} are uniformly bounded, and*

$$v^{\ell}_{r,J}(a) := \frac{\bar{B}}{J^2}\sum_{j=1}^{J}\frac{1-\pi_j}{\pi_j}a_j^2(r_j^{\ell})^2 \longrightarrow v_r^{\ell}(a) < \infty. \tag{OA.1.26}$$

Then

$$\sqrt{\bar{B}}\{\widehat{\Phi}^{\ell}_J(a) - \Phi^{\ell}_J(a)\} \Rightarrow \mathcal{N}\big(0,\, v_r^{\ell}(a) + v^{\ell}_{\mathrm{diag}}(a) + 4(K-1)\tau_1^{\ell}(a)\big)\,. \tag{OA.1.27}$$

The same conclusion holds with $\sqrt{B}$ in place of $\sqrt{\bar{B}}$.

Proof. The residuals $r_j^{\ell} = \vartheta_j - g_j^{\ell}$ are deterministic and uniformly bounded. Together with balanced spending and (OA.1.26), this allows us to apply Lemma 1, giving

$$\sqrt{\bar{B}}\,T^{s}_{J,\ell}(a) \Rightarrow \mathcal{N}\big(0, v_r^{\ell}(a)\big).$$

We now consider the measurement term conditional on $\mathcal{G}_J$. Write

$$\sqrt{\bar{B}}\,T^{m}_{J,\ell}(a) = \sum_{j\in\mathcal{S}_J} Z_{Jj}, \qquad Z_{Jj} := \frac{\sqrt{\bar{B}}}{J}\frac{a_j}{\pi_j}(\widehat{\vartheta}_j - \vartheta_j).$$

Equation (OA.1.3) gives $\mathbb{E}(Z_{Jj} \mid \mathcal{G}_J) = 0$.

Join two selected documents when they share an annotator. Two collections of documents with no edge between them use disjoint sets of annotator blocks. By design invariance, these blocks remain independent conditional on $\mathcal{G}_J$. The resulting graph is therefore a dependency graph for the document contributions. Each document has at most $\bar{m}$ annotators, and each of those annotators evaluates at most $K-1$ other documents, so its maximum degree is at most $\bar{m}(K-1)$.

The bounded kernel gives $|\widehat{\vartheta}_j - \vartheta_j| \le 2C_{\ell}$. Consequently,

$$|Z_{Jj}| \le \frac{2C_{\ell} m_j |a_j|}{\lambda_j \sqrt{\bar{B}}} \le \frac{C_0}{\sqrt{\bar{B}}}, \tag{OA.1.28}$$

where $C_0 < \infty$ is independent of J. The conditional variance of the sum is $v^{\ell}_{\varepsilon,J}(a)$, and Theorem OA.1 gives

$$v^{\ell}_{\varepsilon,J}(a) \xrightarrow{p} v^{\ell}_{\varepsilon}(a) := v^{\ell}_{\mathrm{diag}}(a) + 4(K-1)\tau_1^{\ell}(a) \ge 0.$$

Also, $|\mathcal{S}_J| \le B$, and Lemma 5 gives $B/\bar{B} \xrightarrow{p} 1$.

For fixed $t \in \mathbb{R}$, define the conditional characteristic function

$$\psi_J(t) := \mathbb{E}\Big[e^{it\sqrt{\bar{B}}T^{m}_{J,\ell}(a)} \,\Big|\, \mathcal{G}_J\Big]\,.$$

Suppose first that $v_\varepsilon^\ell(a) > 0$. With probability tending to one, $B \le 2\bar{B}$ and $v_{\varepsilon,J}^\ell(a) \ge v_\varepsilon^\ell(a)/2$. On this event, the sufficient condition in Janson (1988, Theorem 2), using exponent 3, is verified by

$$|\mathcal{S}_J|\{1+\bar{m}(K-1)\}^2 \left\{ \frac{C_0}{\sqrt{\bar{B}}\sqrt{v_{\varepsilon,J}^\ell(a)}} \right\}^3 = O(\bar{B}^{-1/2}) \longrightarrow 0.$$

To apply this result conditionally, take any subsequence. There is a further subsequence along which $B/\bar{B} \to 1$ and $v_{\varepsilon,J}^\ell(a) \to v_\varepsilon^\ell(a)$ almost surely. Along almost every such sequence of graph realizations, the preceding bounds verify Janson's theorem for the conditional distributions. It follows that, along the original sequence,

$$\psi_J(t) \xrightarrow{p} e^{-t^2 v_\varepsilon^\ell(a)/2}.$$

If $v_\varepsilon^\ell(a) = 0$, conditional centering and $|e^{ix} - 1 - ix| \le x^2/2$ give directly

$$|\psi_J(t) - 1| \le \frac{t^2}{2} v_{\varepsilon,J}^\ell(a) \xrightarrow{p} 0.$$

Thus the same conditional characteristic-function convergence holds in both cases. Since the difference is bounded by two,

$$\mathbb{E}\left|\psi_J(t) - e^{-t^2 v_\varepsilon^\ell(a)/2}\right| \longrightarrow 0. \tag{OA.1.29}$$

Finally, the sampling term is $\mathcal{G}_J$-measurable. The conditional characteristic-function argument in the proof of Theorem 1 therefore applies to (OA.1.14), using (OA.1.29) and the sampling limit above. This proves (OA.1.27). Since $B/\bar{B} \xrightarrow{p} 1$, Slutsky's theorem gives the same limit under the $\sqrt{B}$ normalization. □

OA.1.3 Variance estimation and feasible inference

OA.1.3.1 Construction and unbiasedness

We construct a consistent estimator of the variance in Theorem OA.2. The sampling and measurement components require estimates of

$$(r_j^\ell)^2 = (\vartheta_j - g_j^\ell)^2, \qquad \mathrm{Cov}(\widehat{\vartheta}_j, \widehat{\vartheta}_k \mid \mathcal{G}_J),$$

respectively, where $j = k$ includes the within-document variance.

As in the weighted-mean case, the observed squared residual contains an additional annotation-variance term:

$$\mathbb{E}\Big[(\widehat{\vartheta}_j - g_j^\ell)^2 \,\Big|\, \mathcal{G}_J\Big] = (r_j^\ell)^2 + \mathrm{Var}(\widehat{\vartheta}_j \mid \mathcal{G}_J).$$

Because the number of labels per document remains bounded, this additional term again, need not vanish. We therefore first construct unbiased estimates of the conditional variances and covariances. From this, the within-document variance estimate can then supply the correction to the squared residual.

The covariance identity

$$\mathrm{Cov}(\widehat{\vartheta}_j, \widehat{\vartheta}_k \mid \mathcal{G}_J) = \mathbb{E}(\widehat{\vartheta}_j \widehat{\vartheta}_k \mid \mathcal{G}_J) - \vartheta_j \vartheta_k$$

suggests subtracting an unbiased estimate of $\vartheta_j \vartheta_k$ from the observed product $\widehat{\vartheta}_j \widehat{\vartheta}_k$. We obtain this estimate by multiplying kernel evaluations based on disjoint annotator pairs.

For this construction, including the case $j = k$, suppose

$$4 \le m_j \le \bar{m} \qquad \text{whenever } \pi_j > 0. \tag{OA.1.30}$$

For an annotator pair $A = \{i, r\} \subseteq M_j$, write

$$\ell_{j,A} := \ell(Y_{ij}^*, Y_{rj}^*).$$

For selected documents j, k, define

$$\mathcal{D}_{jk} := \{(A, C) : A \subseteq M_j,\ C \subseteq M_k,\ |A| = |C| = 2,\ A \cap C = \varnothing\}. \tag{OA.1.31}$$

This set contains all ordered choices of two disjoint annotator pairs. It is nonempty because, after choosing A, at least two annotators remain in $M_k \setminus A$.

Define the average product

$$P_{jk}^{\perp} := \frac{1}{|\mathcal{D}_{jk}|} \sum_{(A,C)\in\mathcal{D}_{jk}} \ell_{j,A}\ell_{k,C}. \tag{OA.1.32}$$

Conditional on $\mathcal{G}_J$, the two kernel evaluations in each product use disjoint, independent annotator blocks. Consequently,

$$\mathbb{E}(\ell_{j,A}\ell_{k,C} \mid \mathcal{G}_J) = \vartheta_j\vartheta_k, \qquad \mathbb{E}(P_{jk}^{\perp} \mid \mathcal{G}_J) = \vartheta_j\vartheta_k.$$

Our conditional covariance estimator is therefore

$$\widehat{c}_{jk} := \widehat{\vartheta}_j\widehat{\vartheta}_k - P_{jk}^{\perp}. \tag{OA.1.33}$$

For distinct documents, $\widehat{c}_{jk}$ plays the role of $\widehat{\Gamma}_{jk}/(m_j m_k)$ in the weighted-mean variance estimator (4.16). Here, products based on disjoint annotator pairs estimate $\vartheta_j\vartheta_k$ directly, so no additional leave-out correction factor is needed. The resulting estimator accounts for both projection orders in the conditional covariance. For $j = k$, it estimates $\mathrm{Var}(\widehat{\vartheta}_j \mid \mathcal{G}_J)$.

Using $\widehat{c}_{jj}$ to remove the annotation variance, define

$$\widehat{r}_{j,\ell}^{\,2} := (\widehat{\vartheta}_j - g_j^{\ell})^2 - \widehat{c}_{jj}. \tag{OA.1.34}$$

This is the same correction used for weighted means, with $\widehat{c}_{jj}$ replacing the estimated variance of the document mean.

Substituting these corrected residual squares into the document-sampling variance formula gives

$$\widehat{v}_{r,J}^{\ell}(a) := \frac{\bar{B}}{J^2} \sum_{j\in\mathcal{S}_J} \frac{1-\pi_j}{\pi_j^2} a_j^2 \widehat{r}_{j,\ell}^{\,2}. \tag{OA.1.35}$$

For the measurement component, we substitute $\widehat{c}_{jk}$ for each conditional covariance in (OA.1.15). If $N_{jk} = 0$, every annotator pair from M_j is disjoint from every annotator pair from M_k, so

$$P_{jk}^{\perp} = \widehat{\vartheta}_j\widehat{\vartheta}_k, \qquad \widehat{c}_{jk} = 0.$$

Only document pairs sharing an annotator therefore contribute. Define

$$\widehat{v}_{\varepsilon,J}^{\ell}(a) := \frac{\bar{B}}{J^2} \left[\sum_{j\in\mathcal{S}_J} \frac{a_j^2}{\pi_j^2}\widehat{c}_{jj} + \sum_{\substack{j,k\in\mathcal{S}_J \\ j\neq k,\ N_{jk}\geq 1}} \frac{a_j a_k}{\pi_j\pi_k}\widehat{c}_{jk} \right], \tag{OA.1.36}$$

where the second sum is over ordered document pairs. The total variance estimator is

$$\widehat{v}_J^{\ell}(a) := \widehat{v}_{r,J}^{\ell}(a) + \widehat{v}_{\varepsilon,J}^{\ell}(a). \tag{OA.1.37}$$

Proposition OA.1 (Unbiasedness)**.** *Under the maintained assumptions and* (OA.1.30)*,*

$$\mathbb{E}\widehat{v}_{r,J}^{\ell}(a) = v_{r,J}^{\ell}(a), \tag{OA.1.38}$$

$$\mathbb{E}\big[\widehat{v}_{\varepsilon,J}^{\ell}(a) \,\big|\, \mathcal{G}_J\big] = v_{\varepsilon,J}^{\ell}(a). \tag{OA.1.39}$$

Consequently,

$$\mathbb{E}\widehat{v}_J^{\ell}(a) = \bar{B}\,\mathrm{Var}\{\widehat{\Phi}_J^{\ell}(a) - \Phi_J^{\ell}(a)\}. \tag{OA.1.40}$$

Proof. The preceding construction gives, for selected j, k, including $j = k$,

$$\mathbb{E}(\widehat{c}_{jk} \mid \mathcal{G}_J) = \mathrm{Cov}(\widehat{\vartheta}_j, \widehat{\vartheta}_k \mid \mathcal{G}_J). \tag{OA.1.41}$$

The residual correction therefore satisfies

$$\begin{aligned}\mathbb{E}(\widehat{r}^2_{j,\ell} \mid \mathcal{G}_J) &= \mathbb{E}\Big[(\widehat{\vartheta}_j - g^\ell_j)^2 \,\Big|\, \mathcal{G}_J\Big] - \mathbb{E}(\widehat{c}_{jj} \mid \mathcal{G}_J) \\ &= (r^\ell_j)^2. \end{aligned} \tag{OA.1.42}$$

Substituting (OA.1.41) into the measurement-variance estimator proves (OA.1.39). For the sampling component, taking conditional expectations first gives

$$\mathbb{E}\big[\widehat{v}^\ell_{r,J}(a) \mid \mathcal{G}_J\big] = \frac{\bar{B}}{J^2} \sum_{j=1}^{J} \frac{I_j(1-\pi_j)}{\pi_j^2} a_j^2 (r^\ell_j)^2.$$

Taking expectations and using $\mathbb{E}I_j = \pi_j$ proves (OA.1.38). The total-variance identity then follows from the law of total variance and the conditional mean-zero property of the measurement error. □

For the sampling component, define the infeasible estimator as in the main text,

$$\widehat{v}^{\ell,\mathrm{or}}_{r,J}(a) := \frac{\bar{B}}{J^2} \sum_{j=1}^{J} \frac{I_j(1-\pi_j)}{\pi_j^2} a_j^2 (r^\ell_j)^2.$$

The unbiasedness proof gives

$$\mathbb{E}\big[\widehat{v}^\ell_{r,J}(a) \mid \mathcal{G}_J\big] = \widehat{v}^{\ell,\mathrm{or}}_{r,J}(a).$$

The Poisson-sampling calculation in (D.4), with r_j replaced by r^ℓ_j and $\lambda_j \geq c_\lambda$, therefore gives

$$\mathrm{Var}\{\widehat{v}^{\ell,\mathrm{or}}_{r,J}(a)\} = O(\bar{B}^{-1}).$$

It remains to control the annotation noise around this conditional mean. Boundedness of the kernel and frozen predictions implies that $\widehat{r}^2_{j,\ell}$ is uniformly bounded in absolute value. Its coefficient satisfies

$$\frac{\bar{B}}{J^2} \frac{1-\pi_j}{\pi_j^2} a_j^2 = \frac{(1-\pi_j) a_j^2 m_j^2}{\lambda_j^2 \bar{B}} \leq \frac{C_0}{\bar{B}}.$$

With the graph fixed, replacing one annotator's response block changes the corrected residuals of at most K documents. The change in $\widehat{v}^\ell_{r,J}(a)$ is consequently at most $C_0/\bar{B}$.

The annotator-replacement argument in the proof of Proposition 4 therefore applies. Applying the Efron–Stein inequality to the conditional law given $\mathcal{G}_J$, with at most B independent active-annotator blocks, gives

$$\mathrm{Var}\big(\widehat{v}^\ell_{r,J}(a) \mid \mathcal{G}_J\big) \leq \frac{C_0 B}{\bar{B}^2}.$$

Using $\mathbb{E}B = \bar{B}$, unbiasedness, and the law of total variance, we obtain

$$\begin{aligned}\mathbb{E}\big[\{\widehat{v}^\ell_{r,J}(a) - v^\ell_{r,J}(a)\}^2\big] &= \mathbb{E}\big[\mathrm{Var}\{\widehat{v}^\ell_{r,J}(a) \mid \mathcal{G}_J\}\big] + \mathrm{Var}\{\widehat{v}^{\ell,\mathrm{or}}_{r,J}(a)\} \\ &= O(\bar{B}^{-1}).\end{aligned}$$

Thus $\widehat{v}^\ell_{r,J}(a) - v^\ell_{r,J}(a) \xrightarrow{p} 0$.

For the measurement component, unbiasedness gives

$$\mathbb{E}\big[\widehat{v}^\ell_{\varepsilon,J}(a) \mid \mathcal{G}_J\big] = v^\ell_{\varepsilon,J}(a).$$

We therefore control the fluctuation around this conditional mean.

Boundedness of the kernel implies

$$|\widehat{c}_{jk}| \le 2C_\ell^2.$$

Moreover, $\widehat{c}_{jk}$ depends only on the responses of annotators in $M_j \cup M_k$, and it is identically zero when $N_{jk} = 0$. The overlap-neighborhood bound (D.11) therefore applies unchanged: replacing one annotator's response block affects at most K diagonal terms and $2K\bar{m}(K-1)$ ordered off-diagonal terms. The graph and the sets $\mathcal{D}_{jk}$ remain fixed during the replacement.

For both diagonal and off-diagonal terms, the coefficients satisfy

$$\frac{\bar{B}}{J^2}\frac{|a_j a_k|}{\pi_j \pi_k} = \frac{|a_j a_k| m_j m_k}{\lambda_j \lambda_k \bar{B}} \le \frac{C_0}{\bar{B}}.$$

Consequently, replacing one annotator's response block changes $\widehat{v}^{\ell}_{\varepsilon,J}(a)$ by at most $C_0/\bar{B}$. Applying the Efron–Stein inequality to the conditional law given $\mathcal{G}_J$, exactly as for the sampling component, gives

$$\mathrm{Var}\big(\widehat{v}^{\ell}_{\varepsilon,J}(a) \mid \mathcal{G}_J\big) \le \frac{C_0 B}{\bar{B}^2}.$$

Conditional unbiasedness and $\mathbb{E}B = \bar{B}$ now imply

$$\begin{aligned}\mathbb{E}\big[\{\widehat{v}^{\ell}_{\varepsilon,J}(a) - v^{\ell}_{\varepsilon,J}(a)\}^2\big] &= \mathbb{E}\big[\mathrm{Var}\{\widehat{v}^{\ell}_{\varepsilon,J}(a) \mid \mathcal{G}_J\}\big] \\ &\le \frac{C_0}{\bar{B}}.\end{aligned}$$

Thus $\widehat{v}^{\ell}_{\varepsilon,J}(a) - v^{\ell}_{\varepsilon,J}(a) \xrightarrow{p} 0$.

OA.2 Stratified fixed-budget inference for weighted averages

OA.2.1 Design and estimator

Maintain the fixed-corpus framework and the main-text conditions on annotator responses, target weights, and frozen predictions. In this section, we replace Poisson document sampling by Design **(SFB)**.

Partition the corpus into a fixed number H of prespecified strata $\mathcal{U}_{h,J}$. Independently within each stratum, select $\mathcal{S}_{h,J}$ by simple random sampling without replacement, with a fixed sample size $2 \le n_h \le J_h$, where

$$J_h := |\mathcal{U}_{h,J}|, \qquad \rho_h := \frac{J_h}{J}, \qquad \pi_h := \frac{n_h}{J_h}.$$

Each selected document in stratum h receives m_h labels, with $2 \le m_h \le \bar{m}$. The annotation budget is therefore fixed:

$$B_h := n_h m_h, \qquad B := \sum_{h=1}^{H} B_h.$$

Each stratum uses a separate annotator pool. Conditional on the selected documents, draw the allocation graph $A_{h,J}$ uniformly from all simple bipartite graphs with document degree m_h and the capacity-filled annotator degrees specified in the main text, with fixed capacity K. Assume that this graph class is nonempty. Sampling and allocation are independent across strata and independent of the annotator blocks. Write

$$\mathcal{G}_J^{\mathrm{sfb}} := \sigma\big((\mathcal{S}_{h,J}, A_{h,J})_{h=1}^{H}\big).$$

For the asymptotic results, suppose that $B \to \infty$ and $\rho_h \to \rho_h^0 > 0$ for every h. Define

$$\lambda_h := \frac{J\pi_h m_h}{B}, \tag{OA.2.1}$$

and impose the stratum-level analogue of the main-text balanced-spending condition:

$$\inf_{J,h} \lambda_h > 0.$$

Since $B_h/B = \rho_h \lambda_h$, each stratum receives a share of the total budget bounded away from zero. Together with $2 \le m_h \le \bar{m}$, this implies $n_h \asymp B_h \asymp B$.

Estimate the weighted corpus mean $\Phi_J(a)$ by

$$\widehat{\Phi}_J^{\mathrm{sfb}}(a) := \frac{1}{J}\sum_{j=1}^{J} a_j g_j + \sum_{h=1}^{H} \frac{\rho_h}{n_h} \sum_{j \in \mathcal{S}_{h,J}} a_j(\bar{Y}_j - g_j). \tag{OA.2.2}$$

This is the main-text estimator with inclusion probability π_h for documents in stratum h.

To separate document-sampling error from measurement error, write

$$x_j := a_j(\mu_j - g_j), \qquad \bar{x}_h := \frac{1}{J_h} \sum_{j \in \mathcal{U}_{h,J}} x_j.$$

Adding and subtracting the sampled latent means gives

$$\widehat{\Phi}_J^{\mathrm{sfb}}(a) - \Phi_J(a) = T_{s,J}^{\mathrm{sfb}} + T_{m,J}^{\mathrm{sfb}}, \tag{OA.2.3}$$

$$T_{s,J}^{\mathrm{sfb}} := \sum_{h=1}^{H} \rho_h \left\{ \frac{1}{n_h} \sum_{j \in \mathcal{S}_{h,J}} x_j - \bar{x}_h \right\},$$

$$T_{m,J}^{\mathrm{sfb}} := \sum_{h=1}^{H} \frac{\rho_h}{n_h} \sum_{j \in \mathcal{S}_{h,J}} a_j(\bar{Y}_j - \mu_j).$$

The sampling term has mean zero, and the measurement term has conditional mean zero given $\mathcal{G}_J^{\mathrm{sfb}}$. Notice, as commneted in the main text, the unbiasedness proof is still valid here.

OA.2.2 Variance and stabilization

OA.2.2.1 Variance expressions

We first calculate the variances of the two terms in (OA.2.3). For the sampling term, let

$$S_{x,h}^2 := \frac{1}{J_h - 1} \sum_{j \in \mathcal{U}_{h,J}} (x_j - \bar{x}_h)^2.$$

The SRS variance formula (Li and Ding, 2017, Section 2.1), applied within each stratum, together with independence across strata, gives

$$v_{s,J}^{\mathrm{sfb}} := B \operatorname{Var}(T_{s,J}^{\mathrm{sfb}}) = \sum_{h=1}^{H} \frac{\rho_h m_h}{\lambda_h}(1 - \pi_h) S_{x,h}^2. \tag{OA.2.4}$$

The main-text covariance calculation has already conditioned on the realized sample and allocation, so the sample size is fixed in that calculation even under Poisson sampling. Under either design, design exogeneity ensures that the annotator blocks remain independent and retain their original distributions. Covariances across strata are zero because their annotator pools are disjoint. With $m_j = m_h$ within stratum h, the conditional measurement variance is therefore

$$\begin{aligned} v_{\varepsilon,J}^{\mathrm{sfb}} &:= B \operatorname{Var}\left(T_{m,J}^{\mathrm{sfb}} \,\middle|\, \mathcal{G}_J^{\mathrm{sfb}}\right) \\ &= \sum_{h=1}^{H} \frac{\rho_h}{\lambda_h} \left\{ \frac{1}{n_h} \sum_{j \in \mathcal{S}_{h,J}} a_j^2 \nu_j + \frac{1}{B_h} \sum_{\substack{j,k \in \mathcal{S}_{h,J} \\ j \neq k}} N_{jk}^{(h)} a_j a_k \tau_{jk} \right\}. \end{aligned} \tag{OA.2.5}$$

To express the average measurement variance, averaged across sampling and allocation, write

$$\bar{d}_h := \frac{1}{J_h} \sum_{j \in \mathcal{U}_{h,J}} a_j^2 \nu_j, \qquad \tau_h := \frac{1}{J_h(J_h-1)} \sum_{\substack{j,k \in \mathcal{U}_{h,J} \\ j \neq k}} a_j a_k \tau_{jk}. \tag{OA.2.6}$$

Also define

$$\bar{k}^{(2)}_{h,J} := \frac{1}{B_h} \sum_i k_{hi}(k_{hi}-1), \tag{OA.2.7}$$

where k_{hi} is the prescribed degree of annotator i in stratum annotator pool h, and the sum is over that annotator pool.

Under capacity filling within each stratum, every active annotator has degree K, except possibly one partially filled annotator. Thus $\bar{k}^{(2)}_{h,J} = K-1$ whenever B_h is divisible by K. More generally, since $\sum_i k_{hi} = B_h$,

$$0 \leq (K-1) - \bar{k}^{(2)}_{h,J} = \frac{1}{B_h} \sum_i k_{hi}(K - k_{hi}) \leq \frac{K^2}{4B_h}. \tag{OA.2.8}$$

Only the possible partially filled annotator contributes to the last sum, and $k_{hi}(K-k_{hi}) \leq K^2/4$. We retain $\bar{k}^{(2)}_{h,J}$ below to account for this finite-budget correction exactly.

Proposition OA.2 (Variance expression)**.** *Under Design **(SFB)**, the average measurement variance is*

$$\bar{v}^{\mathrm{sfb}}_{\varepsilon,J} := \mathbb{E}\big[v^{\mathrm{sfb}}_{\varepsilon,J}\big] = \sum_{h=1}^{H} \frac{\rho_h}{\lambda_h} \left\{ \bar{d}_h + \bar{k}^{(2)}_{h,J} \tau_h \right\}. \tag{OA.2.9}$$

Consequently,

$$B \operatorname{Var}\Big(\widehat{\Phi}^{\mathrm{sfb}}_J(a) \Big) = v^{\mathrm{sfb}}_{s,J} + \bar{v}^{\mathrm{sfb}}_{\varepsilon,J}. \tag{OA.2.10}$$

Proof. Within stratum h, all selected documents have degree m_h. The uniform allocation law therefore gives the same expected overlap for every pair of distinct selected documents. The total overlap satisfies

$$\sum_{\substack{j,k \in \mathcal{S}_{h,J} \\ j \neq k}} N^{(h)}_{jk} = \sum_i k_{hi}(k_{hi}-1) = B_h \bar{k}^{(2)}_{h,J}.$$

There are $n_h(n_h-1)$ ordered pairs of distinct selected documents. Each pair has the same expected overlap conditional on $\mathcal{S}_{h,J}$, and the sum of these expectations equals the fixed total $B_h \bar{k}^{(2)}_{h,J}$ displayed above. The common expected overlap is therefore this total divided by the number of ordered pairs:

$$\mathbb{E}\Big[N^{(h)}_{jk} \,\Big|\, \mathcal{S}_{h,J} \Big] = \frac{B_h \bar{k}^{(2)}_{h,J}}{n_h(n_h-1)}, \qquad j \neq k. \tag{OA.2.11}$$

We next average over document sampling. Under SRS, document j is selected with probability n_h/J_h. Conditional on its selection, the remaining $n_h - 1$ documents are sampled uniformly from the other $J_h - 1$ documents. Thus, for distinct $j, k \in \mathcal{U}_{h,J}$,

$$\mathbb{P}(j,k \in \mathcal{S}_{h,J}) = \frac{n_h}{J_h} \frac{n_h - 1}{J_h - 1} = \frac{n_h(n_h-1)}{J_h(J_h-1)}.$$

For the within-document contribution,

$$\mathbb{E}\left[\frac{1}{n_h} \sum_{j \in \mathcal{S}_{h,J}} a_j^2 \nu_j \right] = \frac{1}{n_h} \sum_{j \in \mathcal{U}_{h,J}} \frac{n_h}{J_h} a_j^2 \nu_j = \bar{d}_h.$$

For the shared-annotator contribution, first average over allocation using (OA.2.11), and then over document sampling:

$$\begin{aligned}
&\mathbb{E}\left[\frac{1}{B_h}\sum_{\substack{j,k\in\mathcal{S}_{h,J}\\ j\neq k}} N_{jk}^{(h)} a_j a_k \tau_{jk}\right] \\
&\quad= \frac{\bar{k}_{h,J}^{(2)}}{n_h(n_h-1)}\mathbb{E}\left[\sum_{\substack{j,k\in\mathcal{S}_{h,J}\\ j\neq k}} a_j a_k \tau_{jk}\right] \\
&\quad= \frac{\bar{k}_{h,J}^{(2)}}{n_h(n_h-1)}\frac{n_h(n_h-1)}{J_h(J_h-1)}\sum_{\substack{j,k\in\mathcal{U}_{h,J}\\ j\neq k}} a_j a_k \tau_{jk} \\
&\quad= \bar{k}_{h,J}^{(2)}\tau_h.
\end{aligned}$$

Substituting these two expectations into (OA.2.5) gives (OA.2.9).

Finally, conditional on $\mathcal{G}_J^{\mathrm{sfb}}$, the sampling error $T_{s,J}^{\mathrm{sfb}}$ is fixed and the measurement error $T_{m,J}^{\mathrm{sfb}}$ has mean zero. Consequently,

$$\mathbb{E}\big[T_{s,J}^{\mathrm{sfb}} T_{m,J}^{\mathrm{sfb}}\big] = \mathbb{E}\big[T_{s,J}^{\mathrm{sfb}}\mathbb{E}\big[T_{m,J}^{\mathrm{sfb}} \,\big|\, \mathcal{G}_J^{\mathrm{sfb}}\big]\big] = 0,$$

so the two errors are uncorrelated. The same conditional mean-zero property and the law of total variance give

$$B\,\mathrm{Var}(T_{m,J}^{\mathrm{sfb}}) = \mathbb{E}\big[v_{\varepsilon,J}^{\mathrm{sfb}}\big] = \bar{v}_{\varepsilon,J}^{\mathrm{sfb}}.$$

Adding this to the sampling variance $B\,\mathrm{Var}(T_{s,J}^{\mathrm{sfb}}) = v_{s,J}^{\mathrm{sfb}}$ proves (OA.2.10).

□

OA.2.2.2 Variance stabilization

We now show that the conditional measurement variance $v_{\varepsilon,J}^{\mathrm{sfb}}$ concentrates around its expectation $\bar{v}_{\varepsilon,J}^{\mathrm{sfb}}$. The law of total variance separates its mean-square deviation into variation from allocation and variation from document sampling:

$$\begin{aligned}
\mathbb{E}\Big[\big(v_{\varepsilon,J}^{\mathrm{sfb}} - \bar{v}_{\varepsilon,J}^{\mathrm{sfb}}\big)^2\Big] &= \mathbb{E}\big[\mathrm{Var}\big(v_{\varepsilon,J}^{\mathrm{sfb}} \,\big|\, \mathcal{S}_{1,J},\ldots,\mathcal{S}_{H,J}\big)\big] \\
&\quad + \mathrm{Var}\big(\mathbb{E}\big[v_{\varepsilon,J}^{\mathrm{sfb}} \,\big|\, \mathcal{S}_{1,J},\ldots,\mathcal{S}_{H,J}\big]\big)\,.
\end{aligned} \tag{OA.2.12}$$

The first term measures variation across allocations for a fixed selected sample. We control it by applying Lemma OA.2 within each stratum. The second measures variation across selected samples after averaging over allocation. The resulting expression contains within-stratum sample averages of $a_j^2\nu_j$ and pair averages of $a_j a_k \tau_{jk}$. The SRS variance formula controls the sample averages; Lemma OA.6 controls the pair averages. Together, these bounds will show that both terms in (OA.2.12) are $O(B^{-1})$.

We first consider variation from annotator allocation. Conditional on the selected documents, each stratum has the same uniform fixed-degree allocation law covered by the earlier overlap-concentration argument. We therefore apply that argument within each stratum and combine the resulting bounds.

Lemma OA.5 (Concentration over annotator allocation)**.** *Under Design **(SFB)**, for all sufficiently large J,*

$$\mathrm{Var}\big(v_{\varepsilon,J}^{\mathrm{sfb}} \,\big|\, \mathcal{S}_{1,J},\ldots,\mathcal{S}_{H,J}\big) \le \frac{C_0}{B}, \tag{OA.2.13}$$

uniformly over the selected sets.

Proof. Conditional on the selected sets, the within-document contribution in (OA.2.5) is fixed. As the main proof, we can define the shared-annotator contribution in stratum h, define

$$c_{jk}^{(h)} := \frac{\rho_h}{\lambda_h B_h} a_j a_k \tau_{jk} = \frac{a_j a_k \tau_{jk}}{B\lambda_h^2}, \qquad j,k \in \mathcal{S}_{h,J}, \quad j \neq k,$$

where the second equality uses $B_h = B\rho_h\lambda_h$. These coefficients are fixed conditional on the selected set. Boundedness of the weights and response covariances, together with the lower bound on λ_h, gives

$$\max_{j\neq k} |c_{jk}^{(h)}| = O(B^{-1}),$$

uniformly over strata and selected sets.

Within each stratum, the allocation graph has degree sum B_h, document degrees bounded by $\bar{m}$, and annotator degrees bounded by K. Since $B_h \asymp B \to \infty$, Lemma OA.2 applies and gives

$$\mathrm{Var}\left(\sum_{\substack{j,k\in\mathcal{S}_{h,J}\\ j\neq k}} c_{jk}^{(h)} N_{jk}^{(h)} \,\middle|\, \mathcal{S}_{h,J} \right) \leq \frac{C_0 B_h}{B^2}.$$

Allocations are independent across strata conditional on the selected sets. Consequently,

$$\begin{aligned}\mathrm{Var}\big(v_{\varepsilon,J}^{\mathrm{sfb}} \,\big|\, \mathcal{S}_{1,J},\dots,\mathcal{S}_{H,J}\big) &= \sum_{h=1}^{H} \mathrm{Var}\left(\sum_{\substack{j,k\in\mathcal{S}_{h,J}\\ j\neq k}} c_{jk}^{(h)} N_{jk}^{(h)} \,\middle|\, \mathcal{S}_{h,J} \right) \\ &\leq \frac{C_0}{B^2} \sum_{h=1}^{H} B_h = \frac{C_0}{B}.\end{aligned}$$

□

Taking expectations in (OA.2.13) over document sampling bounds the first term in (OA.2.12) by C_0/B.

The preceding allocation-concentration lemma allows us to replace the graph-specific measurement variance by its conditional mean given the selected documents, with a remainder whose mean square vanishes. We make this replacement explicit before controlling the remaining sampling variation.

For each stratum, write

$$\bar{d}_{\mathcal{S}_h} := \frac{1}{n_h} \sum_{j\in\mathcal{S}_{h,J}} a_j^2 \nu_j, \qquad \tau_{\mathcal{S}_h} := \frac{1}{n_h(n_h-1)} \sum_{\substack{j,k\in\mathcal{S}_{h,J}\\ j\neq k}} a_j a_k \tau_{jk}.$$

Define the allocation remainder

$$R_J^{\mathrm{alloc}} := v_{\varepsilon,J}^{\mathrm{sfb}} - \mathbb{E}\big[v_{\varepsilon,J}^{\mathrm{sfb}} \,\big|\, \mathcal{S}_{1,J},\dots,\mathcal{S}_{H,J}\big].$$

By its definition, R_J^{alloc} has conditional mean zero given the selected documents. Lemma OA.5 controls its second moment:

$$\mathbb{E}\big[(R_J^{\mathrm{alloc}})^2\big] = \mathbb{E}\big[\mathrm{Var}\big(v_{\varepsilon,J}^{\mathrm{sfb}} \,\big|\, \mathcal{S}_{1,J},\dots,\mathcal{S}_{H,J}\big)\big] \leq \frac{C_0}{B}. \tag{OA.2.14}$$

Substituting the expected overlaps from (OA.2.11) into (OA.2.5) therefore yields

$$\begin{aligned} v_{\varepsilon,J}^{\mathrm{sfb}} &= \mathbb{E}\big[v_{\varepsilon,J}^{\mathrm{sfb}} \,\big|\, \mathcal{S}_{1,J},\dots,\mathcal{S}_{H,J}\big] + R_J^{\mathrm{alloc}} \\ &= \sum_{h=1}^{H} \frac{\rho_h}{\lambda_h} \left\{ \bar{d}_{\mathcal{S}_h} + \bar{k}_{h,J}^{(2)} \tau_{\mathcal{S}_h} \right\} + R_J^{\mathrm{alloc}}. \end{aligned} \tag{OA.2.15}$$

The allocation remainder is thus already controlled. Comparing the remaining expression with (OA.2.9), we need to control

$$\bar{d}_{\mathcal{S}_h} - \bar{d}_h, \qquad \tau_{\mathcal{S}_h} - \tau_h.$$

For the diagonal term, apply the same SRS variance formula used in (OA.2.4) (Li and Ding, 2017, Section 2.1), now to the fixed values $a_j^2\nu_j$. Since $\mathbb{E}[\bar{d}_{\mathcal{S}_h}] = \bar{d}_h$, this gives

$$\begin{aligned}\mathbb{E}\big[(\bar{d}_{\mathcal{S}_h} - \bar{d}_h)^2\big] &= \left(\frac{1}{n_h} - \frac{1}{J_h}\right)\frac{1}{J_h - 1}\sum_{j\in\mathcal{U}_{h,J}}(a_j^2\nu_j - \bar{d}_h)^2 \\ &\le \frac{C_0}{n_h},\end{aligned}$$

where the inequality follows from uniform boundedness of $a_j^2\nu_j$.

For the off-diagonal term, we follow the same approach as in the main-text stabilization proof: decompose $\tau_{\mathcal{S}_h} - \tau_h$ into a linear term and a quadratic remainder, and bound their second moments separately. In the main text, independence of the centered selection indicators makes the relevant cross-products vanish. Under SRS, those indicators are dependent, so we must obtain the required cancellations differently.

Lemma OA.6 (Concentration of the off-diagonal term under SRS)**.** *Under Design **(SFB)** and the maintained boundedness conditions, consider*

$$\tau_{\mathcal{S}_h} = \frac{1}{n_h(n_h-1)}\sum_{\substack{j,k\in\mathcal{S}_{h,J}\\ j\neq k}} a_j a_k \tau_{jk}, \qquad \tau_h = \frac{1}{J_h(J_h-1)}\sum_{\substack{j,k\in\mathcal{U}_{h,J}\\ j\neq k}} a_j a_k \tau_{jk}.$$

These quantities average the weighted same-annotator covariances over distinct documents in the selected sample and in the entire stratum, respectively. The first is random through document selection; the second is its deterministic population counterpart. For all sufficiently large J, and for each stratum h,

$$\mathbb{E}[\tau_{\mathcal{S}_h}] = \tau_h, \qquad \mathbb{E}\big[(\tau_{\mathcal{S}_h} - \tau_h)^2\big] \le \frac{C_0}{n_h}, \tag{OA.2.16}$$

where C_0 is uniform over strata and population sizes.

Proof. Consider stratum h, with population $\mathcal{U}_{h,J}$ containing J_h documents and selected sample $\mathcal{S}_{h,J}$ containing n_h documents. For distinct documents $j, k \in \mathcal{U}_{h,J}$, write

$$b_{jk} := a_j a_k \tau_{jk}.$$

Thus b_{jk} is the weighted same-annotator covariance between documents j and k. These values are deterministic and symmetric, and the maintained boundedness conditions give $|b_{jk}| \le M$, uniformly over documents, strata, and population sizes.

The sampled off-diagonal quantity and its population counterpart are therefore

$$\tau_{\mathcal{S}_h} = \frac{1}{n_h(n_h-1)}\sum_{j\in\mathcal{S}_{h,J}}\left[\sum_{k\in\mathcal{S}_{h,J}\setminus\{j\}} b_{jk}\right],$$

and

$$\tau_h = \frac{1}{J_h(J_h-1)}\sum_{j\in\mathcal{U}_{h,J}}\left[\sum_{k\in\mathcal{U}_{h,J}\setminus\{j\}} b_{jk}\right].$$

The first expression sums over the selected documents; the second sums over the entire stratum.

Every two distinct documents are selected together with probability $n_h(n_h-1)/\{J_h(J_h-1)\}$. Therefore,

$$\mathbb{E}[\tau_{\mathcal{S}_h}] = \frac{1}{n_h(n_h-1)} \sum_{\substack{j,k \in \mathcal{U}_{h,J} \\ j \neq k}} \frac{n_h(n_h-1)}{J_h(J_h-1)} b_{jk} = \tau_h.$$

Since $J_h \geq n_h \to \infty$, take J sufficiently large that $J_h \geq 4$.

We begin a Hoeffding-type decomposition by constructing the contribution associated with each individual document.

For each document $r \in \mathcal{U}_{h,J}$, define

$$\psi_{h,r} := \frac{1}{J_h - 1} \sum_{k \in \mathcal{U}_{h,J} \setminus \{r\}} b_{rk} - \tau_h.$$

Here document r is held fixed, and the sum averages its weighted covariances with the other documents in the stratum. Thus $\psi_{h,r}$ measures how far this average lies above or below the stratum-wide average τ_h.

This construction follows the same principle as the first projection in a Hoeffding decomposition: hold one document fixed, average over the other document, and subtract the overall mean. Under sampling without replacement, a finite-population correction is needed to obtain the exact linear component; we derive that correction below.

The deviations $\psi_{h,j}$ sum to zero within stratum h. This follows because averaging the document-specific averages over all documents recovers the stratum-wide average τ_h.

Notice, the definition of τ_h gives

$$\sum_{j \in \mathcal{U}_{h,J}} \left[\sum_{k \in \mathcal{U}_{h,J} \setminus \{j\}} b_{jk} \right] = J_h(J_h-1)\tau_h.$$

The double sum contains $J_h(J_h-1)$ terms, whose average is τ_h. Therefore, summing $\psi_{h,j}$ over all J_h documents yields

$$\begin{aligned}
\sum_{j \in \mathcal{U}_{h,J}} \psi_{h,j} &= \frac{1}{J_h - 1} \sum_{j \in \mathcal{U}_{h,J}} \left[\sum_{k \in \mathcal{U}_{h,J} \setminus \{j\}} b_{jk} \right] - J_h \tau_h \\
&= \frac{J_h(J_h-1)\tau_h}{J_h - 1} - J_h \tau_h \\
&= 0.
\end{aligned}$$

We next remove these document-specific contributions from $b_{jk} - \tau_h$. Our aim is to make the remaining terms sum to zero separately for each document. This will provide the cancellations needed to bound the quadratic remainder.

To determine the required adjustment, fix any document j. By the definition of $\psi_{h,j}$,

$$\sum_{k \in \mathcal{U}_{h,J} \setminus \{j\}} (b_{jk} - \tau_h) = (J_h - 1)\psi_{h,j}.$$

Meanwhile, for each fixed document $r \in \mathcal{U}_{h,J}$,

$$\begin{aligned}
\sum_{k \in \mathcal{U}_{h,J} \setminus \{r\}} (\psi_{h,r} + \psi_{h,k}) &= (J_h - 1)\psi_{h,r} + \sum_{k \in \mathcal{U}_{h,J} \setminus \{r\}} \psi_{h,k} \\
&= (J_h - 1)\psi_{h,r} + \left(\sum_{k \in \mathcal{U}_{h,J}} \psi_{h,k} - \psi_{h,r} \right) \\
&= (J_h - 1)\psi_{h,r} - \psi_{h,r} \\
&= (J_h - 2)\psi_{h,r}.
\end{aligned}$$

The fixed value $\psi_{h,r}$ appears $J_h - 1$ times. The other deviations sum to $-\psi_{h,r}$, because their total over the entire stratum is zero.

We next construct the canonical second-order term of the Hoeffding-type decomposition. We subtract the overall mean and the first-order contributions associated with both documents, choosing the correction factor so that the resulting term averages to zero over the other documents whenever one document is held fixed. Define

$$\alpha_{h,J} := \frac{J_h - 1}{J_h - 2}, \qquad b^\circ_{jk} := b_{jk} - \tau_h - \alpha_{h,J}(\psi_{h,j} + \psi_{h,k}).$$

For every fixed document $r \in \mathcal{U}_{h,J}$,

$$\begin{aligned}\sum_{k \in \mathcal{U}_{h,J} \setminus \{r\}} b^\circ_{rk} &= \sum_{k \in \mathcal{U}_{h,J} \setminus \{r\}} (b_{rk} - \tau_h) - \alpha_{h,J} \sum_{k \in \mathcal{U}_{h,J} \setminus \{r\}} (\psi_{h,r} + \psi_{h,k}) \\ &= (J_h - 1)\psi_{h,r} - \alpha_{h,J}(J_h - 2)\psi_{h,r} \\ &= 0.\end{aligned}$$

Since $|b_{jk}| \le M$, the population average satisfies $|\tau_h| \le M$. For each document $r \in \mathcal{U}_{h,J}$, the triangle inequality gives

$$|\psi_{h,r}| \le \frac{1}{J_h - 1} \sum_{k \in \mathcal{U}_{h,J} \setminus \{r\}} |b_{rk}| + |\tau_h| \le M + M = 2M.$$

For all sufficiently large J, we have $J_h \ge 4$, and therefore

$$\alpha_{h,J} = \frac{J_h - 1}{J_h - 2} = 1 + \frac{1}{J_h - 2} \le \frac{3}{2}.$$

Finally, for distinct documents $j, k \in \mathcal{U}_{h,J}$,

$$\begin{aligned}|b^\circ_{jk}| &\le |b_{jk}| + |\tau_h| + \alpha_{h,J}\big(|\psi_{h,j}| + |\psi_{h,k}|\big) \\ &\le M + M + \frac{3}{2}(2M + 2M) \\ &= 8M.\end{aligned}$$

Thus both the document-specific deviations and the canonical second-order terms are uniformly bounded.

We now substitute

$$b_{jk} - \tau_h = \alpha_{h,J}(\psi_{h,j} + \psi_{h,k}) + b^\circ_{jk}$$

into the sampled double sum:

$$\begin{aligned}\tau_{\mathcal{S}_h} - \tau_h &= \frac{1}{n_h(n_h - 1)} \sum_{j \in \mathcal{S}_{h,J}} \sum_{\substack{k \in \mathcal{S}_{h,J} \\ k \ne j}} (b_{jk} - \tau_h) \\ &= \frac{\alpha_{h,J}}{n_h(n_h - 1)} \sum_{j \in \mathcal{S}_{h,J}} \sum_{\substack{k \in \mathcal{S}_{h,J} \\ k \ne j}} (\psi_{h,j} + \psi_{h,k}) + \frac{1}{n_h(n_h - 1)} \sum_{j \in \mathcal{S}_{h,J}} \sum_{\substack{k \in \mathcal{S}_{h,J} \\ k \ne j}} b^\circ_{jk}.\end{aligned}$$

In the first double sum, each selected document's $\psi_{h,j}$ appears $n_h - 1$ times as the first term and $n_h - 1$ times as the second term. Consequently,

$$\tau_{\mathcal{S}_h} - \tau_h = \underbrace{\frac{2\alpha_{h,J}}{n_h} \sum_{j \in \mathcal{S}_{h,J}} \psi_{h,j}}_{L_{\mathcal{S}_{h,J}}} + \underbrace{\frac{1}{n_h(n_h - 1)} \sum_{j \in \mathcal{S}_{h,J}} \sum_{\substack{k \in \mathcal{S}_{h,J} \\ k \ne j}} b^\circ_{jk}}_{R_{\mathcal{S}_{h,J}}}. \tag{OA.2.17}$$

We bound the second moments of the linear term $L_{\mathcal{S}_{h,J}}$ and the quadratic remainder $R_{\mathcal{S}_{h,J}}$ separately.

For the linear term, the ordinary SRS variance formula and $\sum_{j\in\mathcal{U}_{h,J}}\psi_{h,j}=0$ give

$$\mathbb{E}[L^2_{\mathcal{S}_{h,J}}]=\frac{4\alpha^2_{h,J}}{n_h}\left(1-\frac{n_h}{J_h}\right)\frac{1}{J_h-1}\sum_{j\in\mathcal{U}_{h,J}}\psi^2_{h,j}\le\frac{CM^2}{n_h}.$$

For the quadratic remainder, symmetry allows us to write

$$Q_{\mathcal{S}_{h,J}}:=\sum_{\substack{j<k\\ j,k\in\mathcal{S}_{h,J}}}b^\circ_{jk},\qquad R_{\mathcal{S}_{h,J}}=\frac{Q_{\mathcal{S}_{h,J}}}{\binom{n_h}{2}}.$$

To expand its second moment, put

$$A:=\sum_{\substack{j,k\in\mathcal{U}_{h,J}\\ j<k}}(b^\circ_{jk})^2.$$

Among products of two distinct summands in $\sum_{\substack{j,k\in\mathcal{U}_{h,J}\\ j<k}}b^\circ_{jk}$, let W denote the sum of those sharing one document and D the sum of those involving four distinct documents. Each product is counted once in W or D.

The zero sums established above imply

$$0=\sum_{j\in\mathcal{U}_{h,J}}\left(\sum_{k\in\mathcal{U}_{h,J}\setminus\{j\}}b^\circ_{jk}\right)^2=2A+2W,$$

so $W=-A$. They also imply $\sum_{\substack{j,k\in\mathcal{U}_{h,J}\\ j<k}}b^\circ_{jk}=0$, and therefore

$$0=\left(\sum_{\substack{j,k\in\mathcal{U}_{h,J}\\ j<k}}b^\circ_{jk}\right)^2=A+2W+2D,$$

so $D=A/2$. These identities control the combined cross-products even though, under SRS, they do not vanish individually.

For $r=2,3,4$, write

$$p_r:=\frac{\binom{n_h}{r}}{\binom{J_h}{r}},$$

with $\binom{n_h}{r}=0$ when $r>n_h$. This is the probability that any specified r distinct documents are all selected. The squared terms in $Q^2_{\mathcal{S}_{h,J}}$ involve two documents, while the products counted by W and D involve three and four documents, respectively. Consequently,

$$\mathbb{E}[Q^2_{\mathcal{S}_{h,J}}]=p_2A+2p_3W+2p_4D=A(p_2-2p_3+p_4)\le Ap_2,$$

where the last inequality uses $p_4\le p_3$. Since $A\le 64M^2\binom{J_h}{2}$, we obtain

$$\mathbb{E}[R^2_{\mathcal{S}_{h,J}}]\le\frac{Ap_2}{\binom{n_h}{2}^2}\le\frac{64M^2}{\binom{n_h}{2}}\le\frac{CM^2}{n_h^2}.$$

Finally, applying $(x+y)^2\le 2x^2+2y^2$ to (OA.2.17) gives

$$\mathbb{E}\big[(\tau_{\mathcal{S}_h}-\tau_h)^2\big]\le 2\mathbb{E}[L^2_{\mathcal{S}_{h,J}}]+2\mathbb{E}[R^2_{\mathcal{S}_{h,J}}]\le\frac{C_0}{n_h}.$$

□

Combining the preceding allocation and document-sampling bounds in (OA.2.12) yields the following result.

Theorem OA.3 (Variance stabilization)**.** *Under Design **(SFB)** and the maintained boundedness conditions, there exists a constant $C_0 < \infty$, independent of J, such that, for all sufficiently large J,*

$$\mathbb{E}\Big[\big(v_{\varepsilon,J}^{\mathrm{sfb}} - \bar{v}_{\varepsilon,J}^{\mathrm{sfb}}\big)^2\Big] \le \frac{C_0}{B}, \tag{OA.2.18}$$

where $\bar{v}_{\varepsilon,J}^{\mathrm{sfb}}$ is the deterministic expectation given in (OA.2.9)*. Consequently,*

$$v_{\varepsilon,J}^{\mathrm{sfb}} - \bar{v}_{\varepsilon,J}^{\mathrm{sfb}} = O_p(B^{-1/2}).$$

OA.2.3 Central limit theorem

We combine the stratumwise SRS central limit theorem with the conditional Gaussian approximation for the measurement error. Variance stabilization allows these two components to be combined using the same conditioning argument as in the main text.

Theorem OA.4 (Stratified fixed-budget central limit theorem)**.** *Under Design **(SFB)** and the maintained boundedness conditions, define*

$$V_J^{\mathrm{sfb}} := v_{s,J}^{\mathrm{sfb}} + \bar{v}_{\varepsilon,J}^{\mathrm{sfb}}.$$

If

$$\liminf_{J\to\infty} V_J^{\mathrm{sfb}} > 0,$$

then

$$\frac{\sqrt{B}\left\{\widehat{\Phi}_J^{\mathrm{sfb}}(a) - \Phi_J(a)\right\}}{\sqrt{V_J^{\mathrm{sfb}}}} \Rightarrow \mathcal{N}(0,1). \tag{OA.2.19}$$

Proof. The deterministic variance components are bounded. From any subsequence, we can therefore extract a further subsequence along which

$$\frac{\rho_h m_h}{\lambda_h}(1-\pi_h)S_{x,h}^2 \longrightarrow v_{s,h} \quad (1 \le h \le H), \qquad \bar{v}_{\varepsilon,J}^{\mathrm{sfb}} \longrightarrow v_m.$$

All limits below are taken along this further subsequence. Write $v_s := \sum_{h=1}^{H} v_{s,h}$. Then $V_J^{\mathrm{sfb}} \to v_s + v_m > 0$.

For any stratum with $v_{s,h} > 0$, boundedness implies that $S_{x,h}^2$ and $1-\pi_h$ are bounded away from zero. Together with bounded $a_j r_j$ and $n_h \asymp B \to \infty$, this verifies the no-dominance condition of the finite-population CLT (Li and Ding, 2017, Theorem 1). A stratum with $v_{s,h} = 0$ contributes $o_p(1)$ to the $\sqrt{B}$-scaled sampling error by Chebyshev's inequality. Independence across the fixed number of strata therefore gives

$$\sqrt{B}\, T_{s,J}^{\mathrm{sfb}} \Rightarrow \mathcal{N}(0, v_s).$$

Conditional on $\mathcal{G}_J^{\mathrm{sfb}}$, the scaled measurement error $\sqrt{B}\, T_{m,J}^{\mathrm{sfb}}$ is a sum of independent centered annotator contributions. The maintained bounds and the lower bound on λ_h bound each contribution by $C/\sqrt{B}$. There are at most B active annotators, so their conditional third absolute moments sum to $O(B^{-1/2})$. Moreover, Theorem OA.3 gives $v_{\varepsilon,J}^{\mathrm{sfb}} \xrightarrow{p} v_m$. The conditional characteristic-function argument in Lemma 6 therefore yields, for every fixed $t \in \mathbb{R}$,

$$\mathbb{E}\Big[e^{it\sqrt{B}T_{m,J}^{\mathrm{sfb}}} \,\Big|\, \mathcal{G}_J^{\mathrm{sfb}}\Big] \xrightarrow{p} e^{-t^2 v_m/2}.$$

By boundedness, this convergence also holds in L^1.

Since $T_{s,J}^{\mathrm{sfb}}$ is $\mathcal{G}_J^{\mathrm{sfb}}$-measurable, the conditioning argument in the proof of Theorem 1 combines these convergences to give

$$\sqrt{B}\left\{\widehat{\Phi}_J^{\mathrm{sfb}}(a) - \Phi_J(a)\right\} \Rightarrow \mathcal{N}(0, v_s + v_m).$$

Since $V_J^{\mathrm{sfb}} \to v_s + v_m > 0$, Slutsky's theorem gives (OA.2.19) along this further subsequence. Every subsequence thus has a further subsequence with the same standardized limit $\mathcal{N}(0,1)$, which proves (OA.2.19) along the full sequence. □

OA.2.4 Feasible variance estimation

We follow the proof of Proposition 4: first establish concentration of an infeasible estimator using the true residuals, then control the error from replacing it by a feasible estimator.

For the sampling component, the change from Poisson sampling to SRS replaces the weighted sum of squared residuals by a within-stratum sample variance. The corresponding infeasible estimator is

$$s^2_{x,\mathcal{S}_h} := \frac{1}{n_h - 1} \sum_{j \in \mathcal{S}_{h,J}} \left\{ a_j r_j - \frac{1}{n_h} \sum_{k \in \mathcal{S}_{h,J}} a_k r_k \right\}^2 .$$

The SRS sample variance is unbiased for the population variance (Li and Ding, 2017, Section 2.1), so

$$\mathbb{E}[s^2_{x,\mathcal{S}_h}] = S^2_{x,h}.$$

To establish concentration, write

$$s^2_{x,\mathcal{S}_h} = \frac{n_h}{n_h - 1} \left\{ \frac{1}{n_h} \sum_{j \in \mathcal{S}_{h,J}} (a_j r_j)^2 - \left(\frac{1}{n_h} \sum_{j \in \mathcal{S}_{h,J}} a_j r_j \right)^2 \right\}.$$

Applying the same SRS variance formula used in (OA.2.4) to the bounded quantities $a_j r_j$ and $(a_j r_j)^2$ bounds the variance of each sample mean by C_0/n_h. Uniform boundedness also bounds the variance of the squared sample mean by C_0/n_h. Since $n_h/(n_h - 1) \le 2$, it follows that

$$\mathbb{E}\big[(s^2_{x,\mathcal{S}_h} - S^2_{x,h})^2\big] = \mathrm{Var}(s^2_{x,\mathcal{S}_h}) \le \frac{C_0}{n_h}. \tag{OA.2.20}$$

We now construct the feasible estimator using the corrections introduced in the main text. The unbiasedness identities in Lemma 9 remain valid under Design **(SFB)**: their proof conditions on the realized sample and allocation and does not use Poisson document sampling.

Rewrite the infeasible estimator as

$$s^2_{x,\mathcal{S}_h} = \frac{1}{n_h} \sum_{j \in \mathcal{S}_{h,J}} a_j^2 r_j^2 - \frac{1}{n_h(n_h - 1)} \sum_{j \in \mathcal{S}_{h,J}} \sum_{k \in \mathcal{S}_{h,J} \setminus \{j\}} a_j a_k r_j r_k.$$

For the first term, Lemma 9 gives

$$\mathbb{E}\big[\widehat{r}_j^2 \,\big|\, \mathcal{G}_J^{\mathrm{sfb}}\big] = r_j^2.$$

For the second term, the conditional covariance of $\bar{Y}_j$ and $\bar{Y}_k$ is $N_{jk}^{(h)} \tau_{jk}/m_h^2$. Using the same lemma, with $\widehat{\Gamma}_{jk}^{(h)}$ computed within stratum h's annotator pool, therefore gives

$$\mathbb{E}\left[(\bar{Y}_j - g_j)(\bar{Y}_k - g_k) - \frac{\widehat{\Gamma}_{jk}^{(h)}}{m_h^2} \,\middle|\, \mathcal{G}_J^{\mathrm{sfb}} \right] = r_j r_k.$$

Substituting these estimators into the preceding identity, define

$$\begin{aligned} \widehat{S}^2_{x,h} := {} & \frac{1}{n_h} \sum_{j \in \mathcal{S}_{h,J}} a_j^2 \widehat{r}_j^2 \\ & - \frac{1}{n_h(n_h - 1)} \sum_{j \in \mathcal{S}_{h,J}} \sum_{k \in \mathcal{S}_{h,J} \setminus \{j\}} a_j a_k \left\{ (\bar{Y}_j - g_j)(\bar{Y}_k - g_k) - \frac{\widehat{\Gamma}_{jk}^{(h)}}{m_h^2} \right\}. \end{aligned} \tag{OA.2.21}$$

The two local identities imply

$$\mathbb{E}\Big[\widehat{S}^2_{x,h}\,\Big|\,\mathcal{G}_J^{\mathrm{sfb}}\Big] = s^2_{x,\mathcal{S}_h},$$

and hence $\mathbb{E}[\widehat{S}^2_{x,h}] = S^2_{x,h}$.

We now control the error from replacing the infeasible estimator. As in the proof of Proposition 4, hold $\mathcal{G}_J^{\mathrm{sfb}}$ fixed and replace all responses supplied by one annotator in stratum h by an independent copy.

The residual and covariance calculations in that proof show that this replacement changes at most K estimates $\widehat{r}_j^2$ and at most $2Km_h(K-1)$ nonzero estimates $\widehat{\Gamma}^{(h)}_{jk}$. Each change is uniformly bounded. With their coefficients in (OA.2.21), these terms therefore contribute at most C_0/n_h to the change in the estimator.

For the remaining products $(\bar{Y}_j - g_j)(\bar{Y}_k - g_k)$, a change is possible only if the annotator labels at least one of the two documents. Since the annotator labels at most K documents, at most $2K(n_h-1)$ ordered products can change. Each change is uniformly bounded, including when both factors change. Their coefficient $1/\{n_h(n_h-1)\}$, together with bounded weights, therefore bounds their combined change by C_0/n_h.

Combining these changes by the triangle inequality bounds the change in the full estimator $\widehat{S}^2_{x,h}$ by C_0/n_h. Apply the Efron–Stein inequality to the conditional law of the independent annotator-response blocks given $\mathcal{G}_J^{\mathrm{sfb}}$ as before. There are at most $B_h = n_h m_h$ active annotators. Using the conditional unbiasedness established above and using the exact same logic as that in the main text gives

$$\begin{aligned}\mathbb{E}\Big[(\widehat{S}^2_{x,h} - s^2_{x,\mathcal{S}_h})^2\,\Big|\,\mathcal{G}_J^{\mathrm{sfb}}\Big] &= \mathrm{Var}\Big(\widehat{S}^2_{x,h}\,\Big|\,\mathcal{G}_J^{\mathrm{sfb}}\Big)\\ &\le \frac{C_0 B_h}{n_h^2} \le \frac{C_0}{n_h}.\end{aligned} \tag{OA.2.22}$$

Taking expectations and combining this with (OA.2.20), we obtain

$$\begin{aligned}\mathbb{E}\Big[(\widehat{S}^2_{x,h} - S^2_{x,h})^2\Big] &\le 2\mathbb{E}\Big[(\widehat{S}^2_{x,h} - s^2_{x,\mathcal{S}_h})^2\Big]\\ &\quad + 2\mathbb{E}\big[(s^2_{x,\mathcal{S}_h} - S^2_{x,h})^2\big] \le \frac{C_0}{n_h}.\end{aligned} \tag{OA.2.23}$$

The sampling-variance estimator is

$$\widehat{v}^{\,\mathrm{sfb}}_{s,J} := \sum_{h=1}^{H} \frac{\rho_h m_h}{\lambda_h}(1-\pi_h)\widehat{S}^2_{x,h}. \tag{OA.2.24}$$

The preceding unbiasedness calculation and (OA.2.23), together with fixed H, bounded aggregation weights, and $n_h \asymp B$, give

$$\mathbb{E}[\widehat{v}^{\,\mathrm{sfb}}_{s,J}] = v^{\mathrm{sfb}}_{s,J}, \qquad \mathbb{E}\big[(\widehat{v}^{\,\mathrm{sfb}}_{s,J} - v^{\mathrm{sfb}}_{s,J})^2\big] \le \frac{C_0}{B}.$$

For the measurement component, use the main-text estimator within each stratum. Notice, here nothing depends on the sampling framework, so most of the results can be directly re-used:

$$\begin{aligned}\widehat{v}^{\,\mathrm{sfb}}_{\varepsilon,J} := \sum_{h=1}^{H} \frac{\rho_h}{\lambda_h}\Bigg\{ &\frac{1}{n_h}\sum_{j\in\mathcal{S}_{h,J}} a_j^2\widehat{\sigma}_j^2\\ &+ \frac{1}{B_h}\sum_{j\in\mathcal{S}_{h,J}}\sum_{k\in\mathcal{S}_{h,J}\setminus\{j\}} a_j a_k \widehat{\Gamma}^{(h)}_{jk}\Bigg\}.\end{aligned} \tag{OA.2.25}$$

The unbiasedness identities in Lemma 9 and the variance bound in (D.13) were established conditional on the realized graph and do not use Poisson document sampling. They therefore apply under Design **(SFB)**.

Using bounded a_j, the lower bound on λ_h, and $\sum_h n_h \le B$, these results give

$$\mathbb{E}\big[\widehat{v}^{\mathrm{sfb}}_{\varepsilon,J} \,\big|\, \mathcal{G}^{\mathrm{sfb}}_J\big] = v^{\mathrm{sfb}}_{\varepsilon,J},$$
$$\mathrm{Var}\big(\widehat{v}^{\mathrm{sfb}}_{\varepsilon,J} \,\big|\, \mathcal{G}^{\mathrm{sfb}}_J\big) \le \frac{C_0}{B}.$$

Consequently, $\mathbb{E}[\widehat{v}^{\mathrm{sfb}}_{\varepsilon,J}] = \bar{v}^{\mathrm{sfb}}_{\varepsilon,J}$. Combining these identities with Theorem OA.3 yields

$$\mathbb{E}\big[(\widehat{v}^{\mathrm{sfb}}_{\varepsilon,J} - \bar{v}^{\mathrm{sfb}}_{\varepsilon,J})^2\big] = \mathbb{E}\big[\mathrm{Var}\big(\widehat{v}^{\mathrm{sfb}}_{\varepsilon,J} \,\big|\, \mathcal{G}^{\mathrm{sfb}}_J\big)\big] + \mathbb{E}\big[(v^{\mathrm{sfb}}_{\varepsilon,J} - \bar{v}^{\mathrm{sfb}}_{\varepsilon,J})^2\big] \le \frac{C_0}{B}.$$

Define the total variance estimator by

$$\widehat{v}^{\mathrm{sfb}}_J := \widehat{v}^{\mathrm{sfb}}_{s,J} + \widehat{v}^{\mathrm{sfb}}_{\varepsilon,J}. \tag{OA.2.26}$$

The preceding bounds imply

$$\mathbb{E}[\widehat{v}^{\mathrm{sfb}}_J] = V^{\mathrm{sfb}}_J, \tag{OA.2.27}$$
$$\mathbb{E}\big[(\widehat{v}^{\mathrm{sfb}}_J - V^{\mathrm{sfb}}_J)^2\big] \le \frac{C_0}{B}. \tag{OA.2.28}$$

Under the conditions of Theorem OA.4, the usual feasible inference results follow by Slutsky's theorem.

The stratified SRS results above extend to the two-response functionals in Appendix OA.1. For reasons of space, we omit the full derivation and outline the main steps.

First, replace the document mean and its estimator by ϑ_j and $\widehat{\vartheta}_j$, respectively. The sampling arguments apply to the bounded weighted residuals $a_j(\vartheta_j - g^{\ell}_j)$. For the measurement variance, use the covariance identities in Lemma OA.1. The SRS and allocation bounds above control the diagonal and first-projection contributions. The additional off-diagonal contribution arises when two documents share two annotators. It is asymptotically negligible by the counting argument in the proof of Lemma OA.4.

Second, obtain the measurement CLT using the argument in the proof of Theorem OA.2: conditional on the sample and allocation, connect documents that share an annotator. This gives a dependency graph with uniformly bounded degree. Combine the resulting conditional approximation with the SRS sampling CLT and the subsequence argument above to obtain the corresponding standardized CLT whenever the exact scaled total variance is bounded away from zero.

Finally, for feasible variance estimation, use $\widehat{c}_{jk}$ from (OA.1.33) within each stratum, together with the same correction of the within-stratum sample variance used above. This construction requires $m_h \ge 4$. The local centering identities in Proposition OA.1 and the annotator-replacement argument in Appendix OA.1.3 condition on the realized sample and allocation and therefore remain applicable under SRS. Together with the SRS sample-variance bound and variance stabilization, they give consistency of both variance components. The usual feasible inference results follow.

OA.3 Fixed-dimensional joint and smooth-functional inference

This section establishes joint inference for a fixed-dimensional vector containing both one-response and two-response corpus functionals under Poisson document sampling. We follow the argument in Appendix D.3.

OA.3.1 Joint inference for corpus functionals

OA.3.1.1 Moment vector and linear combinations

Let $\ell^{(1)}, \dots, \ell^{(q)}$ be symmetric response kernels and $a^{(1)}, \dots, a^{(q)}$ be target-weight arrays, where $q < \infty$ is fixed. The kernels and weights satisfy the maintained boundedness conditions and are chosen before the study responses are collected. Each coordinate may use a different kernel and different document weights.

For coordinate r, let $\vartheta_j^{(r)}$ and $\widehat{\vartheta}_j^{(r)}$ denote the document target and estimator in (OA.1.1) and (OA.1.2), evaluated using $\ell^{(r)}$. Define

$$\mathbf{\Phi}_J := \big(\Phi_J^{\ell^{(1)}}(a^{(1)}), \ldots, \Phi_J^{\ell^{(q)}}(a^{(q)})\big)^\top, \qquad \Phi_J^{\ell^{(r)}}(a^{(r)}) = \frac{1}{J}\sum_{j=1}^{J} a_j^{(r)}\vartheta_j^{(r)}.$$

For each coordinate r, let $g_j^{(r)}$ be uniformly bounded predictions constructed before the study responses are collected. Using the same document sample and annotator allocation, apply (OA.1.4) coordinatewise:

$$\widehat{\Phi}_{Jr} := \frac{1}{J}\sum_{j=1}^{J} a_j^{(r)} g_j^{(r)} + \frac{1}{J}\sum_{j\in\mathcal{S}_J} \frac{a_j^{(r)}}{\pi_j}\big(\widehat{\vartheta}_j^{(r)} - g_j^{(r)}\big). \tag{OA.3.1}$$

Collect these estimates in $\widehat{\mathbf{\Phi}}_J := (\widehat{\Phi}_{J1}, \ldots, \widehat{\Phi}_{Jq})^\top$.

A one-response coordinate is included by taking

$$\ell^{(r)}(y_1, y_2) = \frac{f^{(r)}(y_1) + f^{(r)}(y_2)}{2},$$

where $f^{(r)}$ is the bounded response function whose mean is of interest. This gives

$$\vartheta_j^{(r)} = \mathbb{E}\Big[f^{(r)}\{Y_j(L)\}\Big], \qquad \widehat{\vartheta}_j^{(r)} = \frac{1}{m_j}\sum_{i\in M_j} f^{(r)}(Y_{ij}^*).$$

In particular, $f^{(r)}(y) = y$ reproduces the weighted mean target and estimator from the main text.

For any $t \in \mathbb{R}^q$ fixed independently of J, define

$$\ell_{j,t}(y_1, y_2) := \sum_{r=1}^{q} t_r a_j^{(r)} \ell^{(r)}(y_1, y_2), \qquad g_j(t) := \sum_{r=1}^{q} t_r a_j^{(r)} g_j^{(r)}.$$

Let $\widehat{\Phi}_J(t)$ be the scalar estimator formed from these kernels and predictions with weights equal to one, and let $\Phi_J(t)$ denote its target. Linearity gives

$$\widehat{\Phi}_J(t) - \Phi_J(t) = t^\top(\widehat{\mathbf{\Phi}}_J - \mathbf{\Phi}_J). \tag{OA.3.2}$$

For each fixed t, the combined kernels remain symmetric and uniformly bounded, and the combined predictions remain uniformly bounded. The preceding scalar results therefore apply under the corresponding sampling- and measurement-variance regularity conditions.

OA.3.1.2 Covariance estimation

As in Appendix D.3, we obtain covariances by estimating the variance of each coordinate and each coordinate sum.

For each fixed $t \in \mathbb{R}^q$, let $\widehat{V}_J(t)$ denote the raw scalar variance estimator in (OA.1.37), evaluated using $\ell_{j,t}$, $g_j(t)$, and weights equal to one. This construction requires $m_j \geq 4$, as in Appendix OA.1.3.

Let e_r denote the rth standard basis vector in $\mathbb{R}^q$. Define

$$\begin{aligned}
\widehat{\Sigma}_{J,rr} &:= \widehat{V}_J(e_r), \\
\widehat{\Sigma}_{J,rs} &:= \frac{1}{2}\Big\{\widehat{V}_J(e_r + e_s) - \widehat{V}_J(e_r) - \widehat{V}_J(e_s)\Big\}, \qquad r \neq s.
\end{aligned} \tag{OA.3.3}$$

All evaluations use the same observed annotations. The covariance estimates therefore account for dependence from shared documents and annotator reuse, including covariances between one-response and two-response coordinates.

OA.3.1.3 Joint limit and covariance consistency

Corollary OA.1 (Joint inference for mean and two-response functionals)**.** *Maintain the conditions of Theorem OA.1, with the limits in* (OA.1.21) *and* (OA.1.22) *required for each fixed linear combination constructed above. Suppose also that the sampling-variance limit* (OA.1.26) *holds for each such combination and that* $m_j \geq 4$ *for every document.*

For each fixed $t \in \mathbb{R}^q$*, let* $V(t)$ *denote the sum of the limiting sampling and measurement variances of the corresponding scalar estimator. Define*

$$\Sigma_{rr} := V(e_r), \qquad \Sigma_{rs} := \frac{1}{2}\{V(e_r + e_s) - V(e_r) - V(e_s)\}, \quad r \neq s. \tag{OA.3.4}$$

Then Σ *is nonnegative semidefinite, and*

$$\sqrt{\bar{B}}\,(\widehat{\boldsymbol{\Phi}}_J - \boldsymbol{\Phi}_J) \Rightarrow \mathcal{N}_q(0, \Sigma), \qquad \widehat{\Sigma}_J \xrightarrow{p} \Sigma. \tag{OA.3.5}$$

Proof. For each fixed t, the linear combination in (OA.3.2) has the scalar prediction-assisted form. Theorem OA.2 therefore gives

$$\sqrt{\bar{B}}\, t^\top(\widehat{\boldsymbol{\Phi}}_J - \boldsymbol{\Phi}_J) \Rightarrow \mathcal{N}(0, V(t)).$$

The sampling and measurement variance expressions are quadratic in t, so their limits satisfy $V(t) = t^\top \Sigma t$. Since $V(t) \geq 0$ for every t, Σ is nonnegative semidefinite. The Cramér–Wold device gives the joint Gaussian limit.

The variance-consistency argument in Appendix OA.1.3, together with Theorem OA.1 and the sampling-variance limits, gives

$$\widehat{V}_J(e_r) \xrightarrow{p} V(e_r), \qquad \widehat{V}_J(e_r + e_s) \xrightarrow{p} V(e_r + e_s).$$

Substitution into (OA.3.3) gives consistency of every covariance entry. Because q is fixed, this proves $\widehat{\Sigma}_J \xrightarrow{p} \Sigma$. □

For stratified SRS, use the stratumwise coordinate estimators and replace $\bar{B}$ by B. Under the corresponding scalar variance-limit and consistency conditions, the same linear-combination and covariance arguments establish joint inference.

Under the conditions stated in Appendix D.4, the smooth-transformation and implicit-parameter results apply directly to the joint limit and covariance estimator established above.

OA.4 Optional model-based inference beyond the realized corpus

OA.4.1 Model and target

Suppose documents are drawn independently from a population in which the mean of μ_j conditional on observed characteristics Z_j is $Z_j^\top \boldsymbol{\beta}_0$. We can conduct inference conditional on the corpus covariates.

$$\mu_j = Z_j^\top \boldsymbol{\beta}_0 + u_j, \qquad \mathbb{E}_P(u_j) = 0, \tag{OA.4.1}$$

where $Z_j \in \mathbb{R}^p$ contains covariates observed for every document, and $\mathbb{E}_P$ denotes expectation under the document model. We condition on these covariates and take p to be fixed. The regressors and model errors are uniformly bounded, and the errors are independent across documents.

Assume

$$Q_J := \frac{1}{J}\sum_{j=1}^{J} Z_j Z_j^\top \longrightarrow Q, \qquad Q \text{ positive definite.}$$

Conditional on the realized corpus, maintain the preceding response and design conditions, with predictions frozen before the final study.

For $r = 1, \ldots, p$, apply (4.1) with weights $a_j^{(r)} = Z_{jr}$, and collect the estimates in $\widehat{\boldsymbol{\Phi}}_J$. Define the realized-corpus regression coefficient and its estimator by

$$\boldsymbol{\beta}_J := Q_J^{-1} \frac{1}{J} \sum_{j=1}^{J} Z_j \mu_j, \qquad \widehat{\boldsymbol{\beta}}_J := Q_J^{-1} \widehat{\boldsymbol{\Phi}}_J. \tag{OA.4.2}$$

The target in this section is the population coefficient $\boldsymbol{\beta}_0$. The estimation error satisfies

$$\widehat{\boldsymbol{\beta}}_J - \boldsymbol{\beta}_0 = (\widehat{\boldsymbol{\beta}}_J - \boldsymbol{\beta}_J) + Q_J^{-1} \frac{1}{J} \sum_{j=1}^{J} Z_j u_j. \tag{OA.4.3}$$

The first term is the fixed-corpus estimation error studied above. The second captures variation of the corpus coefficient under the document model.

The importance of corpus variation depends on the size of the annotation budget relative to the corpus. Independence and boundedness of the model errors imply

$$\sqrt{\bar{B}} \, (\boldsymbol{\beta}_J - \boldsymbol{\beta}_0) = O_p\left(\sqrt{\frac{\bar{B}}{J}} \right).$$

Thus, if $\bar{B}/J \to 0$, the distinction between $\boldsymbol{\beta}_J$ and $\boldsymbol{\beta}_0$ is asymptotically negligible at the inference scale, and the preceding fixed-corpus results also apply to $\boldsymbol{\beta}_0$. If instead $\bar{B}/J \to \kappa > 0$, corpus variation can remain first order. The covariance estimator must then account for this additional source of uncertainty.

OA.4.2 Model-based inference

Let $\widehat{\Sigma}_J$ be the joint covariance estimator in (D.15) for the moment vector used above. To estimate the additional document-model covariance, define

$$\widehat{V}_{P,J}(\boldsymbol{b}) := \frac{\bar{B}}{J^2} \sum_{j \in \mathcal{S}_J} \frac{Z_j Z_j^\top}{\pi_j} \left\{ (\bar{Y}_j - Z_j^\top \boldsymbol{b})^2 - \frac{\widehat{\sigma}_j^2}{m_j} \right\}. \tag{OA.4.4}$$

The subtraction removes annotation noise from the squared regression residual. The total covariance estimator is

$$\widehat{\Sigma}_{\beta,J} := Q_J^{-1} \left\{ \widehat{\Sigma}_J + \widehat{V}_{P,J}(\widehat{\boldsymbol{\beta}}_J) \right\} Q_J^{-\top}. \tag{OA.4.5}$$

Corollary OA.2 (Model-based regression inference)**.** *Maintain the model above and the response, design, allocation, and boundedness conditions of Corollary 2, conditional on the realized corpus. Suppose the variance limits required there hold in probability under the joint document-model and study law, with nonrandom limits defining Σ. Suppose also that $\bar{B}$ is deterministic under the document model and*

$$V_{P,J} := \frac{\bar{B}}{J^2} \sum_{j=1}^{J} Z_j Z_j^\top \mathbb{E}_P(u_j^2) \longrightarrow V_P. \tag{OA.4.6}$$

Then, with $\Sigma_\beta := Q^{-1}(\Sigma + V_P)Q^{-\top}$,

$$\sqrt{\bar{B}} \, (\widehat{\boldsymbol{\beta}}_J - \boldsymbol{\beta}_0) \Rightarrow \mathcal{N}_p(0, \Sigma_\beta), \tag{OA.4.7}$$

$$\widehat{\Sigma}_{\beta,J} \xrightarrow{p} \Sigma_\beta. \tag{OA.4.8}$$

Proof. The preceding joint CLT applies conditional on the realized corpus, with limiting covariance Σ. Its maintained bounds are uniform over admissible corpora, so the conditional characteristic functions converge in probability under the document model.

For the additional term in (OA.4.3), independence and boundedness of u_j and Z_j, together with $\bar{B} \le \bar{m}J$, give the Lindeberg condition. Thus

$$\frac{\sqrt{\bar{B}}}{J}\sum_{j=1}^{J} Z_j u_j \Rightarrow \mathcal{N}_p(0, V_P).$$

The same conditional characteristic-function argument used in the proof of Theorem 1, but this time conditioning on the realized corpus, combines the limits of the two terms in (OA.4.3) after multiplication by $\sqrt{\bar{B}}$. Together with $Q_J \to Q$, this gives the stated Gaussian limit.

For covariance consistency, work first at $\boldsymbol{b} = \boldsymbol{\beta}_0$. Balanced spending and bounded annotation depths imply

$$\frac{\bar{B}}{J^2\pi_j} = \frac{m_j}{J\lambda_j} \le \frac{\bar{m}}{c_\lambda J}.$$

For each selected document j,

$$\mathbb{E}\left[(\bar{Y}_j - Z_j^\top \boldsymbol{\beta}_0)^2 - \frac{\widehat{\sigma}_j^2}{m_j} \,\middle|\, W_{1:J}, \mathcal{G}_J\right] = u_j^2 + \frac{\nu_j}{m_j} - \frac{\nu_j}{m_j} = u_j^2.$$

By linearity of conditional expectation,

$$\mathbb{E}\Big[\widehat{V}_{P,J}(\boldsymbol{\beta}_0) \,\Big|\, W_{1:J}, \mathcal{G}_J\Big] = \frac{\bar{B}}{J^2}\sum_{j\in\mathcal{S}_J} \frac{Z_j Z_j^\top u_j^2}{\pi_j}.$$

By independent Poisson sampling, its mean-squared error around the full-corpus quantity $\bar{B}J^{-2}\sum_j Z_j Z_j^\top u_j^2$ is

$$\frac{\bar{B}^2}{J^4}\sum_{j=1}^{J} \frac{1-\pi_j}{\pi_j}\left\|Z_j Z_j^\top u_j^2\right\|_F^2 \le C\frac{\bar{B}}{J^2},$$

where the inequality follows from balanced spending and the maintained boundedness conditions.

Replacing one annotator's response block affects at most K documents. Bounded responses and the coefficient bound above therefore imply a change of at most C/J in $\widehat{V}_{P,J}(\boldsymbol{\beta}_0)$. With at most B active annotators, the Efron–Stein argument in Proposition 4 gives

$$\mathbb{E}\left[\left\|\widehat{V}_{P,J}(\boldsymbol{\beta}_0) - \frac{\bar{B}}{J^2}\sum_{j\in\mathcal{S}_J}\frac{Z_j Z_j^\top u_j^2}{\pi_j}\right\|_F^2 \,\middle|\, W_{1:J}, \mathcal{G}_J\right] \le \frac{CB}{J^2}.$$

By the law of iterated expectations,

$$\mathbb{E}\left[\left\|\widehat{V}_{P,J}(\boldsymbol{\beta}_0) - \frac{\bar{B}}{J^2}\sum_{j\in\mathcal{S}_J}\frac{Z_j Z_j^\top u_j^2}{\pi_j}\right\|_F^2\right] \le \frac{C}{J^2}\mathbb{E}B = \frac{C\bar{B}}{J^2}.$$

Finally, independence and boundedness of the model errors give

$$\mathbb{E}_P\left\|\frac{\bar{B}}{J^2}\sum_{j=1}^{J} Z_j Z_j^\top\{u_j^2 - \mathbb{E}_P(u_j^2)\}\right\|_F^2 \le C\frac{\bar{B}^2}{J^3}.$$

Decompose

$$\widehat{V}_{P,J}(\boldsymbol{\beta}_0) - V_{P,J} = \underbrace{\left[\widehat{V}_{P,J}(\boldsymbol{\beta}_0) - \frac{\bar{B}}{J^2}\sum_{j\in\mathcal{S}_J}\frac{Z_j Z_j^\top u_j^2}{\pi_j}\right]}_{\text{annotation noise}} + \underbrace{\left[\frac{\bar{B}}{J^2}\sum_{j\in\mathcal{S}_J}\frac{Z_j Z_j^\top u_j^2}{\pi_j} - \frac{\bar{B}}{J^2}\sum_{j=1}^{J} Z_j Z_j^\top u_j^2\right]}_{\text{document selection}} + \underbrace{\left[\frac{\bar{B}}{J^2}\sum_{j=1}^{J} Z_j Z_j^\top u_j^2 - V_{P,J}\right]}_{\text{variation across corpora}}.$$

These three bounds control document selection, annotation noise, and variation across corpora, respectively. They all vanish because $\bar{B} \le \bar{m}J$. Consequently, $\widehat{V}_{P,J}(\boldsymbol{\beta}_0) - V_{P,J} \xrightarrow{p} 0$, and (OA.4.6) gives $\widehat{V}_{P,J}(\boldsymbol{\beta}_0) \xrightarrow{p} V_P$.

The Gaussian limit gives $\widehat{\boldsymbol{\beta}}_J - \boldsymbol{\beta}_0 = o_p(1)$. Expanding the squared residuals, using bounded regressors and $J^{-1}\sum_{j\in\mathcal{S}_J}\pi_j^{-1} = O_p(1)$, shows that

$$\widehat{V}_{P,J}(\widehat{\boldsymbol{\beta}}_J) - \widehat{V}_{P,J}(\boldsymbol{\beta}_0) = o_p(1).$$

Combining this with $\widehat{\Sigma}_J \xrightarrow{p} \Sigma$ and $Q_J \to Q$ proves covariance can be consistently estimated. □

OA.5 Calibration of the annotator-response model

We calibrate the model using all 159,686 comments, 1,598,289 labels and 4,301 annotators in the historical data.

OA.5.1 Comment probabilities

With few ratings per comment, observed toxic-label fractions are noisy and can equal zero or one. We smooth them using a beta-binomial empirical-Bayes model (Piegorsch and Casella, 1996); Humphries et al. (2025, Online Appendix E.4) use a related construction for judges' decision probabilities. If comment j has T_j toxic labels among n_j ratings, its smoothed probability is

$$\mu_j = \frac{T_j + \widehat{\phi}\Lambda(h_j^\top \widehat{b})}{n_j + \widehat{\phi}}, \qquad \Lambda(t) = (1+e^{-t})^{-1}. \tag{OA.5.1}$$

Here h_j contains an intercept, indicators for blocked source, unregistered authorship and user-talk namespace, and $(\text{year}_j - 2010)/5$, in that order. We estimate all five coefficients and the common smoothing weight ϕ together by maximizing the beta-binomial likelihood of the comment counts:

$$\widehat{b} = (-2.90387,\ 1.09658,\ 0.72553,\ 0.57639,\ -0.07674)^\top, \qquad \widehat{\phi} = 1.68912.$$

OA.5.2 Comment characteristics

The annotator model uses the following covariates: abusive-word presence, $-\log(1+\text{word count})$, user-talk namespace, blocked source and unregistered authorship (the latter three are recorded covariates from the original datasets). The abusive-word indicator uses the abusive entries in the base dictionary of Wiegand et al. (2018): 537 distinct words, matched without regard to capitalization and allowing WordNet base forms, such as singular nouns. For each characteristic, we subtract its full-corpus mean and divide by its standard deviation, giving every comment equal weight.

OA.5.3 Annotator parameters

We compare each label with the other annotators' average for the same comment, then average these differences over each annotator's comments:

$$r_{ij} = Y_{ij}^{\text{obs}} - \frac{T_j - Y_{ij}^{\text{obs}}}{n_j - 1}, \qquad m_i = \frac{1}{|\mathcal{J}_i|} \sum_{j \in \mathcal{J}_i} r_{ij},$$

where $\mathcal{J}_i$ is annotator i's set of comments. We fit (σ, c, τ) to products $r_{ij} r_{ik}$ on distinct comments reviewed by the same annotator, and select (q, δ) using the distribution of m_i across annotators.

For each annotator, we form all pairs of distinct comments they reviewed and calculate $r_{ij} r_{ik}$. We group these pairs separately by each comment characteristic. For registration, for example, the three groups contain two comments by registered authors, one by a registered and one by an unregistered author, or two by unregistered authors. Blocked source, user-talk namespace and each vocabulary indicator give three groups in the same way. Length gives ten groups: four within the same length quartile and six between different quartiles.[1]

Within each group, we average all the observed products, giving each product equal weight. An annotator who reviewed more eligible pairs therefore contributes more to that average. This gives 25 observed averages, denoted C_ℓ^{data}.[2]

The model predicts the corresponding averages of $r_{ij} r_{ik}$, denoted C_ℓ^{model}. If another annotator also reviewed both comments, their ratings enter both of the annotator averages subtracted in r_{ij} and r_{ik}. We include this contribution in the model prediction.[3]

We evaluate class shares and intercept differences on a numerical grid, initially covering $q \in [0.005, 0.995]$ and $\delta \in [0, 4]$, with finer spacing around the best-fitting pairs and additional checks at $q = 0.001$, $q = 0.999$ and $\delta = 6$. At every proposed parameter value, we recalculate each comment's intercept so that its response probability, averaged over annotators, remains μ_j. For each pair, we first fit (σ, c, τ) by minimizing

$$L_{\text{cov}} = \sum_{\ell=1}^{25} w_\ell \big(C_\ell^{\text{model}} - C_\ell^{\text{data}}\big)^2. \tag{OA.5.2}$$

We compare the fitted models using the standard deviation, skewness and excess kurtosis of m_i, giving each annotator equal weight. For each model, we generate 200 sets of labels, keeping which annotators review which comments unchanged, and recompute these statistics. Each annotator keeps the same drawn effects across their comments. Let M_h^{data} be an observed statistic and $\overline{M}_h^{\text{model}}$ its average across the generated datasets. We choose the pair minimizing

$$S = \sum_{h=1}^{3} \left(\frac{\overline{M}_h^{\text{model}} - M_h^{\text{data}}}{M_h^{\text{data}}} \right)^2.$$

Thus the three discrepancies receive equal weight relative to their observed values. The selected parameters are

$$(\sigma, c, \tau, q, \delta) = (0.951267,\ -0.137411,\ 0.264315,\ 0.075,\ 2.10).$$

These parameters and the comment probabilities are fixed for all three simulation experiments.

[1] Length quartiles use the full comment corpus. The vocabulary indicators use the Wiegand dictionary and the TOXTRIG "offensive-not-minority" category of Zhou et al. (2021).

[2] In the fitting criterion below, the averages for source, namespace, registration and length each receive total weight 1/5; those for each dictionary receive 1/10. Each total is divided equally among the averages for that characteristic. Each pair contributes to one average for each characteristic.

[3] If s_{jk} annotators reviewed both comments, including annotator i, the model implies

$$\mathbb{E}[r_{ij} r_{ik}] = \left[1 + \frac{s_{jk} - 1}{(n_j - 1)(n_k - 1)}\right] \tau_{jk},$$

where τ_{jk} is the covariance between one annotator's labels for the two comments. For each group, we draw 30,000 combinations of an annotator and two comments they reviewed, uniformly with replacement from that group, and average the predicted products. This sample is held fixed throughout the parameter search.

OA.5.4 Calibration fit

Table OA.5.1 compares observed and model comment ratings, same-annotator dependence and annotator averages. The comparisons use the full historical data.

Table OA.5.1: Calibration fit

Statistic	Observed data	Model
Comment ratings		
Mean toxic-label rate (%)	14.50	14.71
Within-comment pairwise disagreement (%)	13.24	15.02
Comments receiving unanimous ratings (%)	59.51	55.16
Covariance comparisons		
Weighted covariance RMSE	0.000690	
Average same-annotator product $r_{ij}r_{ik}$	0.004951	0.004967
Annotators' average differences m_i		
Standard deviation	0.077424	0.074497
Skewness	1.9558	1.9257
Excess kurtosis	6.4605	6.3138
95th percentile	0.148148	0.144147
99th percentile	0.304670	0.277132

Notes: The first three rows give each comment equal weight. Disagreement averages distinct annotator pairs within each comment; unanimity requires all its labels to agree. Their model values average μ_j, $2\mu_j(1-\mu_j)$ and $\mu_j^{n_j}+(1-\mu_j)^{n_j}$, respectively. RMSE is $\sqrt{L_{\text{cov}}}$. The remaining model entries average another 200 generated datasets, using fresh draws on the historical assignments. Products receive equal annotator–pair weights; the final five rows give annotators equal weight. With $I = 4{,}301$, $\bar{m} = I^{-1}\sum_i m_i$ and $a_r = I^{-1}\sum_i (m_i - \bar{m})^r$, standard deviation is $\sqrt{Ia_2/(I-1)}$, skewness is $a_3/a_2^{3/2}$, and excess kurtosis is $a_4/a_2^2 - 3$.

Using the two observed ratings for each of the 1,000 pilot comments, we use a very naive construction. We construct estimates of annotator-response variance and the squared difference between the machine prediction and the reference-population mean. We fit separate, unweighted ordinary least squares regressions for these two quantities, using an intercept, the five comment characteristics described above, and the machine prediction. We use the fitted regressions to predict both quantities for the remaining comments, setting negative fitted values to zero. These predictions enter the allocation rule to determine comment sampling probabilities and annotation depths. The pilot and resulting allocation are held fixed across Monte Carlo replications.

OA.6 Application: Additional details and robustness checks

OA.6.1 Document sampling

Let ℓ_j be the fixed full-corpus regression weight for the squared-inflation coefficient in the original authors' full-controls specification. Wave one selected documents independently with

$$p_{1j} = \min\{1, \max(p_{\min}, c_1|\ell_j|)\}, \qquad p_{\min} = 0.1\frac{400}{J}, \qquad \sum_j p_{1j} = 400.$$

The multiplier c_1 enforces the expected document budget; the floor preserves coverage for documents with small curvature weights. Wave two blended uniform and curvature-targeted top-up probabilities. Define

$$r_j^C = \min\{1, \max(p_{1j}, c_2|\ell_j|)\}, \quad \sum_j (r_j^C - p_{1j}) = 200, \quad q_j^C = \frac{r_j^C - p_{1j}}{1 - p_{1j}}, \quad q_j^U = \frac{200}{J - 400}, \quad q_{2j} = \tfrac{1}{2}(q_j^C + q_j^U),$$

with $q_j^C = 0$ when $p_{1j} = 1$. The denominator in q_j^U uses the planned first-wave sample size, not its realised size. For independent documentwise bernoulli coins $D_{1j} \sim \text{Bernoulli}(p_{1j})$ and $D_{2j} \sim \text{Bernoulli}(q_{2j})$, actual

wave indicators and union probabilities are

$$I_{1j} = D_{1j}, \qquad I_{2j} = (1 - D_{1j})D_{2j}, \qquad I_j = I_{1j} + I_{2j}, \qquad p_j = p_{1j} + (1 - p_{1j})q_{2j}.$$

Under this fixed probability rule, the I_j are independent across documents. All sampled corrections use p_j. Each selected document receives $m = 2$ ratings, giving expected label budgets 800 and 400 and realised budgets 842 and 384. Conditional on the first draw, the expected second-wave budget was 387.70 labels.

OA.6.2 Annotator allocation

Within each wave, the selected documents were allocated uniformly over simple bipartite graphs with document degree two and capacity-filled annotator loads of at most $K = 8$, allowing one remainder annotator. Annotator pools are disjoint across waves. The final data contain 1,226 ratings on 613 documents from 154 annotators. Sampling variance uses the union probabilities; overlap contributions are calculated within waves and then added.

It should be noted, due to budget disruption, sample collection was forced to be done in two steps and the above proven concentration and inference results do not hold automatically without further considerations. We prove a small extension here to accommodate the top-up structure.

OA.6.3 Two-wave variance concentration

Consider a scalar target $\Phi_J(a) = J^{-1}\sum_j a_j\mu_j$. Maintain boundedness, fixed-predictor and design-exogeneity conditions, and independent annotators from the same reference population. Suppose $\bar{B} = m\sum_j p_j \to \infty$ and, for a fixed constant $c > 0$, combined spending satisfies the one-sided balance condition

$$\frac{Jp_j m}{\bar{B}} \geq c$$

uniformly in j and J. Write $s_{1j} = p_{1j}$, $s_{2j} = (1 - p_{1j})q_{2j}$, $\eta_{hj} = s_{hj}/p_j$, $\bar{B}_h = m\sum_j s_{hj}$, $B_h = m\sum_j I_{hj}$, and $\rho_{h,J} = \bar{B}_h/\bar{B}$, with $\inf_{J,h}\rho_{h,J} > 0$. Using response variance ν_j and shared-annotator covariance τ_{jk}, define the deterministic quantities

$$D_J = \frac{\bar{B}}{mJ^2}\sum_j \frac{a_j^2\nu_j}{p_j}, \qquad T_{h,J} = \frac{1}{J^2}\sum_{j\neq k} a_j a_k \eta_{hj}\eta_{hk}\tau_{jk}, \qquad \bar{v}_{\varepsilon,J} = D_J + (K-1)\sum_{h=1}^{2}\rho_{h,J}^{-1}T_{h,J}.$$

Then the graph-conditional measurement variance satisfies

$$v_{\varepsilon,J} := \bar{B}\,\mathrm{Var}(T_J^m \mid \mathcal{G}_J), \qquad v_{\varepsilon,J} - \bar{v}_{\varepsilon,J} = O_p(\bar{B}^{-1/2}). \tag{OA.6.1}$$

Proof. The diagonal term concentrates around D_J by independent poisson sampling. Within wave h, the indicators are independent across documents and $\mathrm{Var}\{(I_{hj} - s_{hj})/p_j\} \leq 1/p_j$. The pair expansion in (C.23), with these centred variables, therefore gives

$$U_{h,J} := \frac{1}{J^2}\sum_{j\neq k}\frac{I_{hj}I_{hk}}{p_j p_k}a_j a_k\tau_{jk} = T_{h,J} + O_p(\bar{B}^{-1/2}).$$

Conditional on the wave sample, Lemma OA.2 bounds the variance of the centred graph contribution by $O_p(\bar{B}^{-1})$: its pair coefficients are $\bar{B}a_j a_k\tau_{jk}/(J^2 p_j p_k m^2) = O(\bar{B}^{-1})$. The uniform-graph mean (C.15) makes the remaining contribution $(K-1)(\bar{B}/B_h)U_{h,J} + O_p(\bar{B}^{-1})$. Since $B_h/\bar{B}_h = 1 + O_p(\bar{B}^{-1/2})$, summing over the two disjoint annotator pools proves (OA.6.1).

OA.6.4 Feasible inference

This adapts Online Appendix OA.2's deterministic-centering and concentration argument, using the Bernoulli pair calculation in place of its SRS calculation. Let $v_{r,J}$ be (4.4) evaluated with p_j, and set

$V_J = v_{r,J} + \bar{v}_{\varepsilon,J}$. If $\liminf_J V_J > 0$, Proposition 4 and (OA.6.1) give $\widehat{v}_J - V_J = o_p(1)$ for the normalized variance estimator using within-wave overlap. Every subsequence has a further subsequence along which the bounded deterministic sampling and measurement variance centres converge. Equation (OA.6.1) and Theorem 1's conditional characteristic-function argument apply there, proving for the full sequence

$$\frac{\sqrt{B}\{\widehat{\Phi}_J(a) - \Phi_J(a)\}}{\sqrt{\widehat{v}_J}} \;\Rightarrow\; N(0,1).$$

OA.6.5 Alternative targets

Table OA.6.1 reports six additional fits. The original-scale linear model retains speaker effects. Binary logit projects $\mathbf{1}\{Y > 3\}$ using Bernoulli quasi-likelihood; two-annotator averages may equal 0, 1/2 or 1 (Papke and Wooldridge, 1996). The cumulative model sums the two Bernoulli losses for $\mathbf{1}\{Y \geq 3\}$ and $\mathbf{1}\{Y > 3\}$, with common slopes and separate threshold intercepts. Notice that, this is an all-threshold composite projection, not an ordinary ordered-logit category likelihood construction which is outside the scope of this paper (Pedregosa et al., 2017). Combining cumulative logit fits also has an econometric precedent in Baetschmann et al. (2015). Both nonlinear models retain all ten economic terms but omit speaker effects. Their comparisons therefore change the conditioning specification as well as the response target. We use Appendix D.4's implicit-map covariance, including cross-threshold covariances. Astra supplies the corresponding probabilities; the authors' continuous predictor is rounded as $\lfloor g_j^{\text{dict}} + 0.5 \rfloor$ only for nonlinear fits and then thresholded.

Table OA.6.1: Turning points under alternative targets

Target	Auxiliary predictor	Assisted turning point	90% delta-method interval	Predictor-only turning point
Linear 1–5	GPT-6 Astra	1.524	[1.071, 1.977]	1.305
	Authors' score	1.658	[0.995, 2.321]	1.727
Binary logit	GPT-6 Astra	1.502	[1.045, 1.958]	1.238
	Authors' score	1.764	[1.277, 2.250]	1.776
Cumulative logit	GPT-6 Astra	1.599	[1.229, 1.968]	1.231
	Authors' score	1.929	[1.169, 2.690]	1.627

Notes: Same corpus, ratings and inclusion probabilities as the main table. Turning points are in inflation percentage units; in nonlinear models they are minima of the fitted quadratic index. Intervals are nominal pointwise 90% delta-method intervals.

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